GSSSB · STAFF NURSE EXAM

Pharmacology for Nurses

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



10 Chapters | 3D Coloured Diagrams | Pharmacokinetics · Dose calculation | All drug groups & side effects | Antidotes · High-alert drugs | Exam Points (PYQ style)

Hinglish explanation + English exam terms



Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

1 **General Pharmacology**
Definitions, sources, drug nomenclature, routes, pharmacokinetics (ADME, half-life, bioavailability), pharmacodynamics, therapeutic index, adverse drug reactions, drug interactions, drug laws & schedules

2 **Drug Administration, Calculations & Nurse's Responsibilities**
Rights of medication administration, prescription abbreviations, storage, high-alert drugs, dose calculation formulas, IV flow rate, paediatric dosing, medication errors, special routes

3 **Autonomic Nervous System Drugs**
Neurotransmitters & receptors, cholinergic drugs & anticholinesterases, organophosphate poisoning, anticholinergics (atropine), adrenergic agonists (adrenaline, NA, dopamine, dobutamine, salbutamol), alpha & beta blockers

4 **Cardiovascular, Renal & Blood Drugs**
Antihypertensives, antianginals (nitrates), heart failure & digoxin, antiarrhythmics, diuretics, anticoagulants, antiplatelets, thrombolytics, hypolipidaemics, haematinics, IV fluids, electrolytes & blood transfusion

5 **Respiratory & Gastrointestinal Drugs**
Bronchodilators, corticosteroids, antileukotrienes, theophylline, cough remedies, antihistamines; antacids, H2 blockers, PPIs, H. pylori, antiemetics, laxatives, antidiarrhoeals, drugs for liver & IBD

6 **CNS Drugs – Analgesics, Antiepileptics, Anaesthetics & Others**
Opioids & WHO pain ladder, NSAIDs, paracetamol toxicity, antiepileptics & status epilepticus, antiparkinson drugs, sedative-hypnotics, general & local anaesthetics, neuromuscular blockers, malignant hyperthermia, migraine, psychotropics summary

7 **Antimicrobial Drugs**
Principles, penicillins, cephalosporins, carbapenems, aminoglycosides, macrolides, tetracyclines, fluoroquinolones, sulfonamides, metronidazole, vancomycin, anti-TB, anti-leprosy, antifungal, antiviral, antiretroviral, antimalarial, anthelmintic drugs

8 **Endocrine Drugs & Hormones**
Insulin types & administration, hypoglycaemia, oral antidiabetics, thyroid & antithyroid drugs, corticosteroids, sex hormones & contraceptives, uterine drugs, pituitary hormones, drugs for osteoporosis & gout

9 **Anticancer Drugs, Immunopharmacology, Antidotes & Special Populations**
Chemotherapy classes & toxicities, safe handling, immunosuppressants, vaccines & sera, vitamins, poisoning antidotes master table, drugs in pregnancy/ lactation/ children/ elderly/ renal-hepatic disease, drug interactions

10 **Quick Revision – Master Tables & 150 One-Liners**
Drug suffixes, drugs of choice, famous side effects, drugs with narrow therapeutic index, high-alert drugs, monitoring parameters, colour of urine, 150 rapid-fire one-liners and exam strategy

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 General Pharmacology

1.1 Definitions

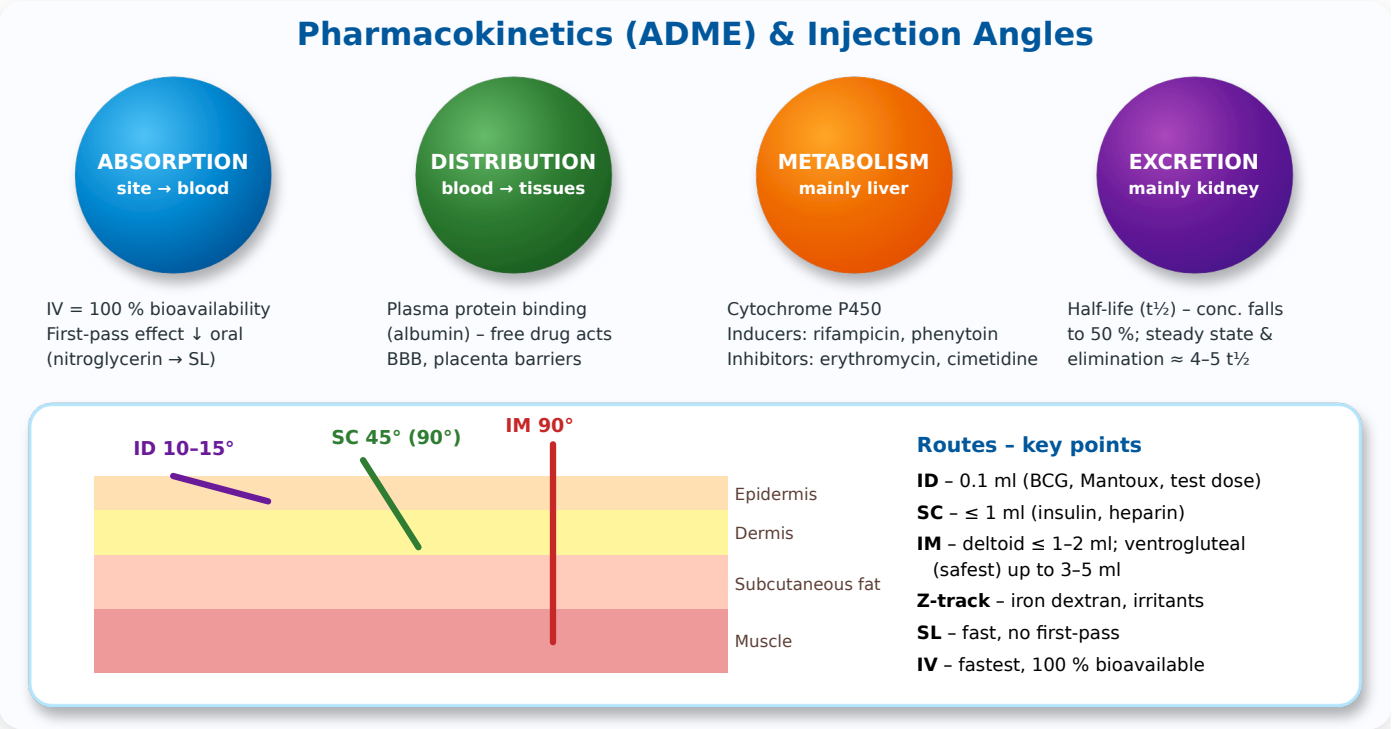
Term	Meaning
Pharmacology	Science of drugs – their sources, properties, actions & uses
Drug (WHO)	Any substance used to **modify or explore physiological systems or pathological states** for the benefit of the recipient
Pharmacokinetics	**What the body does to the drug** – absorption, distribution, metabolism, excretion (**ADME**)
Pharmacodynamics	**What the drug does to the body** – mechanism & effects
Pharmacotherapeutics	Use of drugs in treatment
Pharmacy	Art & science of preparing & dispensing drugs
Toxicology	Study of poisons
Pharmacovigilance	Detection, assessment & prevention of **adverse drug reactions (ADRs)**
Pharmacogenomics	Genetic basis of variation in drug response
Posology	Study of **dosage**
Chemotherapy	Drugs against microbes/ cancer cells
Placebo	Inert substance (**dummy medication**) – psychological effect
Orphan drug	Drug for rare disease
Prodrug	Inactive until metabolised (e.g. **levodopa, enalapril, prednisone**)

- **Father of pharmacology – Oswald Schmiedeberg**; **father of Indian pharmacology – Col. R. N. Chopra**.
- **Sources** – plants (**morphine – opium poppy, atropine – belladonna, digoxin – foxglove (Digitalis lanata), quinine – cinchona, reserpine – Rauwolfia**), animals (insulin earlier, heparin), minerals (iron, $MgSO_4$), microbes (**penicillin – Penicillium notatum, Fleming 1928**), synthetic, **biotechnology (recombinant insulin, vaccines)**.
- **Drug names** – **chemical**, **generic/ non-proprietary (official – e.g. paracetamol)**, **brand/ proprietary (trade – e.g. Crocin)**. **Prescribe by generic name** (cheaper – Jan Aushadhi).
- **Official reference** – **Indian Pharmacopoeia (IP)** – first 1955; by **Indian Pharmacopoeia Commission, Ghaziabad**.

1.2 Routes of Administration

Route	Advantages	Disadvantages
Oral (PO)	**Safest, most convenient, economical**	Slow, **first-pass metabolism**, GI irritation, not in vomiting/ unconscious

Route	Advantages	Disadvantages
Sublingual (SL)/ buccal	**Rapid, bypasses first-pass liver metabolism** (nitroglycerin, buprenorphine)	Only small potent drugs
Rectal	Vomiting/ unconscious patients; partly avoids first-pass	Irregular absorption
Intravenous (IV)	**Fastest onset, 100 % bioavailability**, large volumes, irritants (diluted)	Most dangerous (no recall), phlebitis, embolism, infection
Intramuscular (IM)	Depot preparations, moderate volume	Painful, nerve injury, abscess
Subcutaneous (SC)	Slow sustained absorption (insulin, heparin), self-administration	Small volumes (≤ 1–2 ml)
Intradermal (ID)	**BCG, Mantoux, sensitivity (test) doses** – 0.1 ml at 10–15°	–
Inhalation	Rapid local action (asthma), anaesthetics	Technique-dependent
Transdermal (patches – nitroglycerin, fentanyl, nicotine, hyoscine)	Steady levels, avoids first-pass	Skin irritation
Topical, intrathecal, intra-articular, intraosseous, epidural	Local/ special	–



ADME processes & injection angles/ volumes

1.3 Pharmacokinetics (ADME)

- **Absorption** – movement from site of administration into blood; mostly **passive diffusion**; factors – lipid solubility, ionisation (**weak acids absorbed in stomach – aspirin; weak bases in intestine**), surface area (small intestine – largest), blood flow, food, formulation, **first-pass (presystemic)**

metabolism in gut wall & liver (**high in oral nitroglycerin, lignocaine, propranolol, morphine, testosterone**).

- **Bioavailability** – fraction of drug reaching systemic circulation unchanged; **IV = 100 %**.
- **Distribution** – **volume of distribution (Vd)**; **plasma protein binding (albumin – acidic drugs: warfarin, phenytoin, sulfonamides)** – bound drug inactive (reservoir); displacement → toxicity (**sulfonamides displace bilirubin → kernicterus in neonates**); barriers – **blood-brain barrier, placental barrier** (lipid-soluble drugs cross).
- **Metabolism (biotransformation)** – mainly **liver**; **Phase I** (oxidation, reduction, hydrolysis – **cytochrome P450**), **Phase II** (conjugation – glucuronidation, acetylation (**INH – slow & fast acetylators**), sulphation).
- **Enzyme inducers** (↓ effect of other drugs) – **rifampicin, phenytoin, carbamazepine, phenobarbitone, griseofulvin, chronic alcohol, smoking** (e.g. rifampicin → **OCP failure**).
- **Enzyme inhibitors** (↑ toxicity) – **erythromycin, clarithromycin, ketoconazole, cimetidine, ciprofloxacin, valproate, isoniazid, grapefruit juice, acute alcohol**.
- **Excretion** – mainly **kidney (urine)**; also bile/ faeces, lungs (anaesthetics, alcohol), sweat, saliva, **breast milk** (relevant in lactation); **alkalinisation of urine ↑ excretion of acidic drugs (aspirin, phenobarbitone poisoning)**.
- **Kinetics** – **first-order** (constant **fraction** eliminated per unit time – most drugs) vs **zero-order (saturation)** (constant **amount** – **phenytoin, alcohol, high-dose aspirin, theophylline**) → small dose ↑ causes big level ↑.
- **Plasma half-life (t½)** – time for plasma concentration to fall to **half**; **steady state reached after ~4–5 half-lives**; drug **eliminated (~97 %) after 4–5 half-lives**; **loading dose** to reach target level quickly (long t½ – digoxin, amiodarone); **maintenance dose** to keep it.
- **Therapeutic drug monitoring (TDM)** – narrow therapeutic index drugs – **digoxin, lithium, phenytoin, theophylline, aminoglycosides, vancomycin, cyclosporine, warfarin (INR)**.

1.4 Pharmacodynamics

- **Receptor** – macromolecule to which drug binds to produce effect; types – **G-protein coupled (most common)**, ion channels, enzyme-linked, nuclear (steroids, thyroid).
- **Agonist** (binds & activates – affinity + intrinsic activity), **antagonist** (binds, no activity – blocks agonist; **competitive** – surmountable (atropine), **non-competitive**), **partial agonist** (buprenorphine, aripiprazole), **inverse agonist**.
- **Potency** (dose needed) vs **efficacy** (maximum effect – more important clinically).
- **Therapeutic index (TI) = LD_{50} / ED_{50}** (or TD_{50}/ED_{50}) – **margin of safety**; **narrow TI → TDM needed**.
- **Synergism** – combined effect > sum (**sulfamethoxazole + trimethoprim**; additive – sum); **potentiation** (one drug ↑ effect of other – **levodopa + carbidopa**, **probenecid + penicillin**); **antagonism** – physical, chemical (**protamine–heparin**, chelators), physiological (**adrenaline vs histamine**), receptor.
- **Tolerance** (↑ dose needed – opioids, nitrates), **tachyphylaxis** (rapid tolerance – ephedrine, nitrates), **cross-tolerance**, **dependence**.

1.5 Adverse Drug Reactions (ADR)

Type	Meaning	Example
Side effect	Unwanted but **expected** pharmacological effect at therapeutic dose	Dry mouth with atropine, sedation with antihistamines
Toxic effect	Due to overdose/ accumulation	Paracetamol hepatotoxicity, digoxin toxicity
Type A (Augmented)	**Dose-related, predictable, common (~80 %)**	Hypoglycaemia with insulin, bleeding with warfarin
Type B (Bizarre)	**Not dose-related, unpredictable** – allergy, idiosyncrasy	Anaphylaxis to penicillin
Idiosyncrasy	Genetically determined abnormal reaction	**Haemolysis with primaquine in G6PD deficiency**, succinylcholine apnoea (pseudocholinesterase deficiency), **malignant hyperthermia**
Drug allergy (hypersensitivity)	Immunological	**Type I (anaphylaxis – IgE)**, II (cytotoxic – haemolytic anaemia), III (immune complex – serum sickness), IV (delayed – contact dermatitis)
Iatrogenic	Physician/ treatment induced	Drug-induced Parkinsonism
Teratogenic	Fetal malformation	**Thalidomide – phocomelia**, warfarin, isotretinoin, valproate
Carcinogenic/ mutagenic	Cancer	Anticancer drugs, oestrogens
Withdrawal/ rebound	On stopping	**Clonidine – rebound HTN**, β-blockers – angina, steroids – adrenal crisis
Drug dependence	–	Opioids, BZD
Photosensitivity	–	**Tetracyclines (doxycycline), sulfonamides, chlorpromazine, amiodarone**
Cumulation	Slow elimination	Digoxin

- **Anaphylaxis** – most serious type I reaction – **adrenaline 0.5 mg IM (0.5 ml of 1:1000) – drug of choice**, repeat 5–15 min; O_2, IV fluids, antihistamine, hydrocortisone (adjuncts). **Test dose** before penicillin/ ASV (as per protocol); **observe 30 min**.
- **SJS/ TEN** – sulfonamides, carbamazepine, phenytoin, lamotrigine, allopurinol, nevirapine.
- **Pharmacovigilance Programme of India (PvPI) – 2010** (IPC, Ghaziabad) – **"Suspected ADR Reporting Form"** – **any health professional (including nurses) can report**; helpline **1800-180-3024**; **ADR monitoring centres (AMCs)**.

1.6 Drug Laws & Schedules

Law/ Schedule	Key point
Drugs & Cosmetics Act 1940, Rules 1945	Regulates import, manufacture, sale; **CDSCO** – national regulator; **DCGI** heads
Pharmacy Act 1948	Pharmacist registration; **Pharmacy Council of India**

Law/ Schedule	Key point
NDPS Act 1985	Narcotic drugs & psychotropic substances
Schedule G	"Caution – to be taken under medical supervision" (e.g. antihistamines, hypoglycaemics)
Schedule H	**Prescription-only** drugs ("Rx" – red label)
Schedule H1 (2013)	**3rd/ 4th-gen antibiotics, anti-TB, habit-forming drugs** – separate **register maintained for 3 years**
Schedule X	**Narcotics & psychotropics** (e.g. certain sedatives) – **prescription in duplicate**, records kept 2 yrs, special licence
Schedule C/ C1	Biological products (vaccines, sera, insulin)
Schedule M	Good Manufacturing Practices (GMP)
Schedule Y	Clinical trials (now New Drugs & Clinical Trials Rules 2019)
Schedule K	Exempted drugs
Drugs & Magic Remedies Act 1954	Prohibits misleading advertisements
Drug Price Control Order (DPCO 2013)/ NPPA	Price regulation

- **Essential medicines** – WHO concept (**1977**); **National List of Essential Medicines (NLEM) 2022 – 384 medicines**; **Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP, 2008)** – generic medicines at Jan Aushadhi Kendras.
- **Over-the-counter (OTC)** – no prescription (paracetamol, ORS, antacids).

PYQ style

- What body does to drug – **pharmacokinetics**; drug does to body – **pharmacodynamics**.
- Study of dosage – **posology**.
- Fastest route – **IV**; safest & most common – **oral**.
- SL nitroglycerin avoids – **first-pass metabolism**.
- Bioavailability of IV drug – **100 %**.
- Main site of drug metabolism – **liver**; excretion – **kidney**.
- Enzyme inducer – **rifampicin**; inhibitor – **erythromycin/ cimetidine**.
- Rifampicin + OCP – **contraceptive failure**.
- Zero-order kinetics – **phenytoin, alcohol**.
- Steady state reached after – **4–5 half-lives**.
- Therapeutic index – **LD_{50}/ ED_{50}**; narrow TI – **digoxin, lithium**.
- Sulfonamides in neonates – **kernicterus**.
- Dose-related predictable ADR – **Type A**; allergy – **Type B**.
- Primaquine haemolysis in G6PD – **idiosyncrasy**.
- Thalidomide – **phocomelia (teratogenic)**.
- Anaphylaxis drug of choice – **adrenaline IM 0.5 mg (1:1000)**.
- Prescription-only drugs – **Schedule H**; antibiotics register – **H1**; narcotics – **X**.
- ADR reporting in India – **PvPI (2010)**.
- NLEM 2022 – **384 medicines**.
- Indian Pharmacopoeia Commission – **Ghaziabad**.

2 Drug Administration, Calculations & Nurse's Responsibilities

2.1 Rights of Medication Administration

- **Classic 5 rights** – **Right patient, right drug, right dose, right route, right time**.
- **Extended (10 rights)** – + **right documentation, right to refuse, right assessment, right education (patient), right evaluation/ response**; some add right reason, right form.
- **3 checks** – when **taking from shelf/ cabinet**, when **preparing**, and **before administering (at bedside)**.
- **Two identifiers** – name + ID number/ DOB (**never bed number**).
- **Do NOT give a drug prepared by someone else**; **don't leave medicines at bedside**; **document immediately after giving** (never before).
- **Verbal/ telephone orders** – only in emergency; **read back**, get signed within 24 hrs.

2.2 Prescription Abbreviations

Abbreviation	Latin	Meaning
OD	omni die	Once daily
BD / BID	bis in die	Twice daily
TDS / TID	ter die sumendum	Three times daily
QID / QDS	quater in die	Four times daily
HS	hora somni	At bedtime
SOS	si opus sit	If necessary (once)
PRN	pro re nata	As needed (repeatable)
STAT	statim	**Immediately (once)**
AC / PC	ante cibum / post cibum	Before / after meals
PO	per os	By mouth
NPO / NBM	nil per os	Nothing by mouth
Q4H / Q6H	quaque 4 hora	Every 4 / 6 hours
Rx	recipe	Take (prescription)
Gtt	guttae	Drops
Tab / Cap / Syr / Inj	–	Tablet / capsule / syrup / injection
OS / OD / OU (eye)	oculus sinister/ dexter/ uterque	Left / right / both eyes (avoid – error prone)

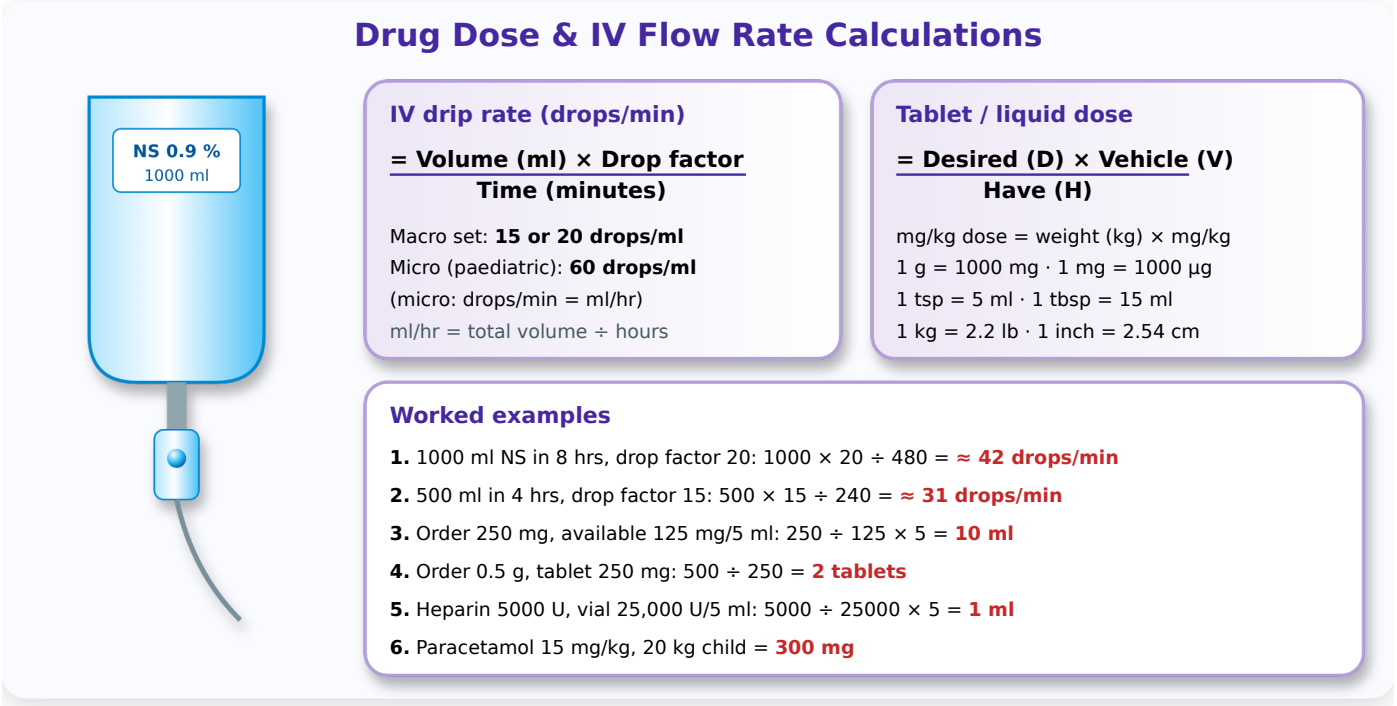
- **"Do not use" list (ISMP)** – U (write "units"), IU, QD, QOD, trailing zero (1.0 mg → write 1 mg), **always leading zero (0.5 mg, not .5 mg)**, MS/ MSO_4.

2.3 Storage of Drugs

- **Cool, dry, dark place**; **original labelled containers**; **check expiry** ("first expiry, first out"); **schedule X/ narcotics in double-lock cupboard** with register & count every shift.
- **Refrigerate (2–8 °C)** – **insulin (unopened), vaccines, oxytocin, methylergometrine, carboprost, anti-D**, **suxamethonium**, some eye drops (as per label); lyophilised ASV can be kept at room temperature.
- **Light-sensitive (protect/ amber)** – **sodium nitroprusside (cover infusion), nitroglycerin (dark glass), frusemide inj, adrenaline, methylergometrine, amphotericin B, vitamin K, chlorpromazine, ciprofloxacin infusion**.
- **Nitroglycerin tablets** – **original dark glass bottle, tightly capped**, replace periodically (loses potency).
- **Multi-dose vials** – date & time of opening; discard as per policy (usually 28 days).

2.4 High-Alert Medications (ISMP)

- **Insulin**, **heparin & anticoagulants**, **concentrated KCl / potassium phosphate**, **opioids**, **sedatives (IV midazolam)**, **neuromuscular blockers**, **chemotherapy**, **hypertonic saline (3 %)**, **magnesium sulphate**, **digoxin IV**, **vasopressors (adrenaline, noradrenaline)**, **thrombolytics**, oral methotrexate (weekly).
- **Independent double check by 2 nurses**, standardised concentrations, infusion pumps, separate storage, **"LASA" (look-alike sound-alike)** caution, **tall-man lettering** (e.g. hydrOXYzine vs hydrALAzine).
- **KCl – NEVER IV push/ bolus (cardiac arrest)**; always diluted; max **10 mEq/hr peripheral** (up to 20 mEq/hr central with monitoring); **≤ 40 mEq/L** in peripheral line; check **urine output** before giving.



IV drip rate & dose calculation formulas with worked examples

2.5 Dose Calculations

- **Basic formula** – **Dose = (Desired ÷ Have) × Vehicle**.

- **Weight-based** – **Dose (mg) = weight (kg) × mg/kg**.
- **IV flow (drops/min) = Volume (ml) × Drop factor (drops/ml) ÷ Time (min)**.
- **Macro-drip – 15 or 20 drops/ml** (adult); **micro-drip – 60 drops/ml** (paediatric) → **drops/min = ml/hr**.
- **ml/hr** (infusion pump) = **total volume ÷ total hours**.
- **Infusion time (hrs)** = volume ÷ ml per hour.
- **Dosage by concentration** – **% solution = g per 100 ml** (e.g. **5 % dextrose = 5 g/100 ml = 50 g/L**; **0.9 % NaCl = 9 g/L**; **10 % calcium gluconate 10 ml = 1 g**); **ratio solutions – 1 : 1000 adrenaline = 1 mg/ml**; **1 : 10,000 = 0.1 mg/ml**.
- **Insulin** – **U-100 = 100 units/ml** (insulin syringe 1 ml = 100 units); **U-40 = 40 units/ml** – use matching syringe.
- **Heparin** – units/ml; **infusion rate (units/hr)**.
- **mcg/kg/min** (vasopressors) – **ml/hr = (dose mcg/kg/min × weight × 60) ÷ concentration (mcg/ml)**.
- **Conversions** – **1 g = 1000 mg; 1 mg = 1000 µg; 1 L = 1000 ml; 1 kg = 2.2 lb; 1 teaspoon = 5 ml; 1 tablespoon = 15 ml; 1 oz = 30 ml; 1 grain ≈ 60–65 mg; °F = (°C × 9/5) + 32**.

Worked examples

Problem	Answer
1000 ml NS over 8 hrs, drop factor 20	1000 × 20 ÷ 480 = **~42 drops/min**
500 ml over 4 hrs, drop factor 15	500 × 15 ÷ 240 = **~31 drops/min**
100 ml antibiotic over 30 min, micro set (60)	100 × 60 ÷ 30 = **200 drops/min** (or 200 ml/hr pump)
Order amoxicillin 250 mg; syrup 125 mg/5 ml	250/125 × 5 = **10 ml**
Order 0.5 g; tablet 250 mg	500/250 = **2 tablets**
Order 75 mg pethidine; vial 100 mg/2 ml	75/100 × 2 = **1.5 ml**
Heparin 5000 U; vial 25,000 U/5 ml	5000/25,000 × 5 = **1 ml**
Gentamicin 5 mg/kg/day for 60 kg	**300 mg/day**
3 L in 24 hrs by pump	3000/24 = **125 ml/hr**

Paediatric dose rules

- **Young's rule** – Child dose = **Age (yrs) ÷ (Age + 12) × adult dose** (1–12 yrs).
- **Clark's rule** – **Weight (lb) ÷ 150 × adult dose**.
- **Fried's rule** – **Age (months) ÷ 150 × adult dose** (infants).
- **BSA rule (most accurate)** – **BSA (m^2) ÷ 1.73 × adult dose**; **Mosteller BSA = √(height cm × weight kg / 3600)**.

2.6 Special Administration Points

Route/ form	Points
Oral	Upright position; **don't crush enteric-coated (EC), sustained-release (SR/ ER/ XL) tablets**; liquid iron via straw; **check swallowing (psychiatric)**; give antacids separately (chelation)
Sublingual	Under tongue, don't swallow/ chew, no water

Route/ form	Points
IM	**Ventrogluteal (safest – no major nerves/ vessels)**, vastus lateralis (infants), deltoid (≤ 1 ml), dorsogluteal (**sciatic nerve risk – upper outer quadrant**); **Z-track** for iron dextran, hydroxyzine; aspirate (not needed for vaccines in deltoid/ VL per current guidance); needle 21–23 G
SC	45–90°; **don't aspirate or massage after heparin/ insulin** (haematoma)
ID	10–15°, bevel up, wheal formed, don't massage
IV	Check patency, **compatibility**, rate; **flush between drugs**; never mix in blood; watch for **extravasation/ infiltration, phlebitis**
Eye drops	Lower conjunctival sac, **press nasolacrimal duct (inner canthus) 1–2 min**; drops before ointment; 5 min gap between drops
Ear drops	Warm to body temp; **adult – pinna up & back**; **child < 3 yrs – down & back**
Nasal drops	Head tilted back
Inhaler (MDI)	Shake, exhale, inhale slowly while pressing, **hold breath 10 sec**; spacer; **wait 1 min between puffs**; **bronchodilator before steroid**; **rinse mouth after ICS**
Vaginal	Lying, after bath, applicator; at bedtime
Rectal suppository	**Left lateral (Sims')**, insert beyond internal sphincter (~10 cm adult)
Transdermal patch	Rotate sites, remove old patch, hairless clean skin, don't cut (fentanyl)

2.7 Medication Errors

- Types – wrong patient, drug, dose, route, time, omission, wrong technique, **LASA**, expired drug.
- **If an error occurs** – **assess & monitor patient first**, **inform doctor/ nurse in-charge immediately**, give antidote/ corrective care, **document facts (incident report – not in patient's chart as "incident report")**, disclose, analyse (system issue – **root cause analysis**), non-punitive culture.
- **Prevention** – rights & checks, clear orders, calculation double-check, **barcode**, avoid distractions, patient involvement.

PYQ style

- STAT – **immediately**; HS – **at bedtime**; SOS – **if necessary once**; PRN – **as needed**.
- TDS – **three times a day**; AC – **before meals**.
- Number of checks before giving drug – **3**.
- Patient identification – **2 identifiers (not bed number)**.
- Documentation – **after giving**.
- KCl – **never IV push**; max peripheral rate – **10 mEq/hr**.
- High-alert drugs – **insulin, heparin, KCl, opioids** – **double check**.
- Drip rate formula – **volume × drop factor ÷ time (min)**.
- Micro-drip – **60 drops/ml**.
- 1000 ml in 8 hrs (DF 20) – **~42 drops/min**.
- 1 : 1000 adrenaline – **1 mg/ml**.
- U-100 insulin – **100 units/ml**.
- 5 % dextrose – **5 g per 100 ml**.
- 1 teaspoon – **5 ml**.
- Safest IM site – **ventrogluteal**; Z-track – **iron dextran**.
- Enteric-coated tablets – **don't crush**.
- Eye drops – **press inner canthus**.
- MDI – **hold breath 10 sec; rinse mouth after steroid**.
- Medication error – first action – **assess the patient**.

3 Autonomic Nervous System Drugs

3.1 ANS Basics

Autonomic Nervous System – Effects & Key Drugs

SYMPATHETIC – "Fight or Flight"

Effects (noradrenaline)
↑ HR & BP (β1, α1) · bronchodilation (β2)
Pupil dilation – mydriasis (α1)
↓ GI motility · urinary retention · ↑ glucose

Agonists (sympathomimetics)
Adrenaline – anaphylaxis, cardiac arrest
Noradrenaline – septic shock (vasopressor)
Dopamine / dobutamine – shock, HF
Salbutamol – β2 – asthma
Phenylephrine – α1 – decongestant

Blockers (sympatholytics)
Propranolol – non-selective β (avoid asthma)
Atenolol, metoprolol – β1 selective
Prazosin – α1 (first-dose hypotension)
Labetalol – α + β (pregnancy HTN)

β-blockers mask hypoglycaemia signs in diabetics

PARASYMPATHETIC – "Rest & Digest"

Effects (acetylcholine – muscarinic)
↓ HR · bronchoconstriction · ↑ secretions
Pupil constriction – miosis
↑ GI motility · urination (SLUDGE)

Cholinergic agonists / anticholinesterases
Pilocarpine – glaucoma (miosis)
Neostigmine – myasthenia, reverse NM block
Pyridostigmine – myasthenia gravis
Bethanechol – urinary retention
Physostigmine – atropine poisoning antidote

Anticholinergics (antimuscarinic)
Atropine – bradycardia, OP poisoning, pre-anaesthetic (↓ secretions)
Ipratropium – COPD · **hyoscine** – colic
Glycopyrrolate – pre-op secretions

Atropine toxicity: dry as bone, red as beet, hot as hare

Sympathetic vs parasympathetic effects & key drugs

	Sympathetic (adrenergic)	Parasympathetic (cholinergic)
Neurotransmitter	**Noradrenaline** (post-ganglionic)	**Acetylcholine**
Heart	↑ rate & force (**β1**)	↓ rate
Bronchi	**Dilate (β2)**	Constrict, ↑ secretion
Pupil	**Dilate – mydriasis (α1)**	**Constrict – miosis**
BP/ vessels	↑ (α1 vasoconstriction)	–
GI	↓ motility	↑ motility & secretions
Bladder	Retention (relax detrusor)	Contraction (urination)
Glucose	↑ (glycogenolysis)	–
Salivary	Thick scanty	Copious watery

- **Receptors** – **α1** (vasoconstriction, mydriasis), **α2** (↓ NA release – clonidine), **β1** (heart), **β2** (bronchi, uterus relax, glycogenolysis), **β3** (lipolysis); cholinergic – **muscarinic (M)** & **nicotinic (Nn – ganglia, Nm – skeletal muscle)**.

3.2 Cholinergic Drugs (Parasympathomimetics)

Drug	Use
Acetylcholine	Rarely (eye surgery – rapid miosis)

Drug	Use
Bethanechol	**Post-operative urinary retention**, paralytic ileus
Pilocarpine	**Glaucoma (miosis ↑ aqueous outflow)**, xerostomia; reverses mydriasis
Neostigmine (anticholinesterase – quaternary, doesn't cross BBB)	**Myasthenia gravis**, **reversal of non-depolarising neuromuscular block** (with atropine/ glycopyrrolate), post-op ileus/ retention, cobra bite
Pyridostigmine	**Myasthenia gravis (drug of choice – longer acting)**
Edrophonium	**Tensilon test** – diagnosis of MG, differentiate myasthenic vs cholinergic crisis
Physostigmine (crosses BBB)	**Atropine/ anticholinergic poisoning antidote**, glaucoma
Donepezil, rivastigmine, galantamine	**Alzheimer's disease**

- **Cholinergic side effects/ toxicity ("SLUDGE/ DUMBELS")** – **S**alivation, **L**acrimation, **U**rination, **D**iarrhoea/ **D**efecation, **GI** cramps, **E**mesis; **bradycardia, bronchospasm/ bronchorrhoea, miosis**, sweating, muscle fasciculations, weakness.
- **Myasthenic crisis** (under-medication – weakness improves with edrophonium) vs **cholinergic crisis** (overdose – worsens; Rx atropine).

Organophosphate (OP) poisoning

- **Irreversible anticholinesterases** – insecticides (**parathion, malathion, chlorpyrifos**), nerve gases; **very common suicidal poisoning in rural India (farmers)**.
- **Features** – **muscarinic (SLUDGE, bradycardia, bronchorrhoea, pin-point pupils)**, **nicotinic (fasciculations, weakness, paralysis – respiratory failure)**, CNS (confusion, seizures, coma); **↓ serum/ RBC cholinesterase**; **kerosene-like smell**.
- **Treatment** – **remove contaminated clothes, wash skin with soap & water** (staff wear gloves), airway, O_2, **ATROPINE (antidote of choice) – 1.8–3 mg IV bolus, double dose every 5 min until "atropinisation"** (**clear chest/ dry secretions (main endpoint), HR > 80, SBP > 80, dry axillae, pupils no longer pin-point**), then infusion; **Pralidoxime (2-PAM – oxime) – cholinesterase reactivator** (give early, before "ageing"); intubation/ ventilation; **intermediate syndrome** (24–96 hrs – proximal & respiratory weakness).
- **Carbamates** (reversible – propoxur, carbaryl) – atropine; oximes not needed generally.

3.3 Anticholinergic (Antimuscarinic) Drugs

Drug	Uses
Atropine (from **Atropa belladonna**)	**Pre-anaesthetic (↓ secretions, prevent vagal bradycardia)**, **symptomatic sinus bradycardia/ heart block (0.5–1 mg IV; ACLS 1 mg every 3–5 min, max 3 mg)**, **OP poisoning**, mushroom poisoning, mydriatic/ cycloplegic (eye exam – refraction in children), with neostigmine
Glycopyrrolate (doesn't cross BBB)	Pre-anaesthetic, with neostigmine
Hyoscine (scopolamine)	**Motion sickness (transdermal patch)**; **hyoscine butylbromide (Buscopan) – abdominal colic**

Drug	Uses
Ipratropium, tiotropium (inhaled)	**COPD, asthma**
Tropicamide, cyclopentolate, homatropine	Mydriatic eye drops (shorter-acting)
Trihexyphenidyl (benzhexol), benztropine	**Parkinsonism, drug-induced EPS**
Oxybutynin, tolterodine	**Overactive bladder/ urge incontinence**, enuresis
Dicyclomine	Abdominal/ menstrual colic, infant colic

- **Anticholinergic side effects/ toxicity** – **dry mouth (most common), blurred vision (cycloplegia), photophobia, mydriasis, tachycardia, urinary retention, constipation, hot dry flushed skin (hyperthermia – esp. children), confusion/ delirium, hallucinations**.
- **Mnemonic** – "**Dry as a bone, red as a beet, hot as a hare, blind as a bat, mad as a hatter, full as a flask (bladder)**".
- **Antidote – physostigmine**; supportive (cooling).
- **Contraindications** – **narrow-angle glaucoma (↑ IOP)**, **BPH/ urinary retention**, paralytic ileus, tachyarrhythmia, myasthenia.
- **Datura poisoning** – anticholinergic (seeds) – same features.

3.4 Adrenergic Agonists (Sympathomimetics)

Drug	Receptor	Key uses & points
Adrenaline (epinephrine)	α, β1, β2	**Anaphylaxis – DOC (0.5 mg IM 1:1000, thigh)**; **cardiac arrest (1 mg IV 1:10,000 every 3–5 min)**; with local anaesthetics (↓ absorption, prolongs action – **not in fingers, toes, ears, nose, penis – gangrene**); bronchospasm; haemostasis (topical)
Noradrenaline (norepinephrine)	α1 > β1	**Vasopressor of choice in septic shock**; **central line preferred**; extravasation → necrosis → **phentolamine** local infiltration
Dopamine	Dose-dependent: **low (1–5 µg/kg/min) – D1 (renal/ mesenteric dilation)**, **medium (5–10) – β1 (inotropic)**, **high (> 10) – α1 (vasoconstriction)**	Cardiogenic/ septic shock, bradycardia
Dobutamine	**β1 (inotrope)**	**Cardiogenic shock, acute heart failure**, stress echo
Isoprenaline	β1 + β2	Heart block, bradycardia (rare)
Phenylephrine	α1	Hypotension during anaesthesia, **nasal decongestant**, mydriatic
Ephedrine	Mixed (indirect)	Hypotension in spinal anaesthesia; tachyphylaxis

Drug	Receptor	Key uses & points
Salbutamol (albuterol), terbutaline	**β2**	**Asthma (reliever – nebulised/ MDI)**, **tocolytic (terbutaline/ ritodrine)**, **hyperkalaemia** (shifts K into cells); side effects – **tremor (most common), tachycardia, palpitations, hypokalaemia**, hyperglycaemia
Salmeterol, formoterol	Long-acting β2 (LABA)	**Maintenance** with ICS (never alone in asthma)
Clonidine	**α2 agonist (central)**	HTN, opioid/ alcohol withdrawal, ADHD; **abrupt stop → rebound hypertension**; sedation, dry mouth
Methyldopa	α2 (central)	**HTN in pregnancy**; haemolytic anaemia, sedation
Midodrine	α1	Orthostatic hypotension

3.5 Adrenergic Blockers

Alpha blockers

Drug	Use / point
Prazosin, terazosin, doxazosin (α1)	HTN, **BPH**, scorpion sting (prazosin); **"first-dose phenomenon" – postural hypotension/ syncope → give first dose at bedtime, lying down**
Tamsulosin (α1A – prostate selective)	**BPH** (less hypotension); retrograde ejaculation
Phentolamine, phenoxybenzamine (non-selective)	**Pheochromocytoma**, extravasation of noradrenaline (phentolamine)

Beta blockers ("-olol")

Type	Drugs
Non-selective (β1 + β2)	**Propranolol** (first; lipid-soluble – CNS), timolol (glaucoma eye drops), nadolol, sotalol
Cardioselective (β1)	**Atenolol, metoprolol, bisoprolol**, esmolol (ultra-short IV), nebivolol
With α-blocking	**Labetalol** (pregnancy HTN, hypertensive emergency), **carvedilol** (heart failure)

- **Uses** – **hypertension, angina, post-MI (↓ mortality), arrhythmias (AF rate control), heart failure (carvedilol, metoprolol, bisoprolol)**, **thyrotoxicosis (symptom control)**, **migraine prophylaxis**, **essential tremor**, **anxiety (performance)**, **glaucoma (timolol)**, portal hypertension (propranolol – varices), hypertrophic cardiomyopathy.
- **Side effects** – **bradycardia, heart block, hypotension**, **bronchospasm (non-selective – contraindicated in asthma)**, **masks hypoglycaemia symptoms (tachycardia, tremor) in diabetics** (sweating persists), fatigue, cold extremities, **sexual dysfunction**, nightmares (propranolol), **worsens Raynaud's**; **abrupt withdrawal → rebound angina/ MI → taper**.
- **Contraindications** – **asthma/ COPD (non-selective)**, **bradycardia (< 50–60)**, **heart block**, decompensated HF, cardiogenic shock.

- **Nursing** – **check apical pulse & BP before giving – hold if HR < 60 or SBP < 90–100** (as per order), teach not to stop abruptly, monitor glucose in diabetics.
- **Beta-blocker overdose antidote – glucagon**.

PYQ style

- Parasympathetic neurotransmitter – **acetylcholine**.
- Pupil – sympathetic **dilation**, parasympathetic **constriction**.
- Post-op urinary retention – **bethanechol**.
- Glaucoma (miotic) – **pilocarpine**; β-blocker eye drop – **timolol**.
- Myasthenia gravis DOC – **pyridostigmine**; diagnosis – **edrophonium (Tensilon)**.
- Reversal of NM block – **neostigmine + atropine/ glycopyrrolate**.
- Atropine poisoning antidote – **physostigmine**.
- OP poisoning antidote – **atropine**; enzyme reactivator – **pralidoxime**.
- Atropinisation endpoint – **dry secretions/ clear chest**.
- OP poisoning pupils – **pin-point**; atropine – **dilated**.
- Atropine contraindicated in – **narrow-angle glaucoma, BPH**.
- Motion sickness – **hyoscine (scopolamine)**.
- Anaphylaxis DOC – **adrenaline 0.5 mg IM (1:1000)**.
- Cardiac arrest adrenaline – **1 mg IV (1:10,000)**.
- Adrenaline with LA – **not in fingers/ toes/ penis**.
- Septic shock vasopressor – **noradrenaline**; extravasation antidote – **phentolamine**.
- Low-dose dopamine – **renal vasodilation**; cardiogenic shock inotrope – **dobutamine**.
- Salbutamol side effect – **tremor, tachycardia, hypokalaemia**.
- Clonidine abrupt stop – **rebound hypertension**.
- Prazosin – **first-dose hypotension (give at bedtime)**.
- Non-selective β-blocker contraindicated in – **asthma**.
- β-blockers mask – **hypoglycaemia** symptoms.
- Hold β-blocker if pulse – **< 60/min**.
- β-blocker overdose antidote – **glucagon**.

4 Cardiovascular, Renal & Blood Drugs

4.1 Antihypertensives

Class	Drugs	Key side effects / points
ACE inhibitors ("-pril")	**Enalapril, captopril, ramipril, lisinopril**	**Dry cough (bradykinin) – most common**, **angioedema**, **hyperkalaemia**, **first-dose hypotension**, ↑ creatinine, **teratogenic (contraindicated in pregnancy)**, bilateral renal artery stenosis; **renoprotective in diabetics**, HF, post-MI
ARBs ("-sartan")	**Losartan, telmisartan, olmesartan**	Like ACEI but **no cough**; hyperkalaemia; contraindicated in pregnancy
Calcium channel blockers	**Amlodipine, nifedipine** (dihydropyridines), **verapamil, diltiazem** (non-DHP)	**Ankle oedema, headache, flushing, reflex tachycardia** (DHP); **bradycardia, constipation (verapamil)**, heart block; **grapefruit ↑ levels**
Thiazide diuretics	**Hydrochlorothiazide, chlorthalidone**, indapamide	First-line HTN; hypokalaemia, hyperglycaemia, hyperuricaemia (gout), hypercalcaemia, hyponatraemia
β-blockers	Atenolol, metoprolol, **labetalol**	See ANS chapter
α1-blockers	Prazosin	First-dose hypotension
Central α2 agonists	**Clonidine, methyldopa**	Clonidine – rebound HTN; **methyldopa – safe in pregnancy**
Vasodilators	**Hydralazine** (pregnancy, lupus-like syndrome), **minoxidil** (hirsutism), **sodium nitroprusside** (**hypertensive emergency – IV; protect from light (wrap in foil); cyanide/ thiocyanate toxicity – max 72 hrs**)	–

- **Hypertensive emergency** – **IV labetalol, nicardipine, sodium nitroprusside, esmolol**; **reduce MAP by ≤ 25 % in first hour** (avoid precipitous fall → stroke).
- **Pregnancy HTN** – **labetalol, nifedipine, methyldopa (hydralazine IV)**; **avoid ACEI/ ARB**.
- **Nursing** – check **BP before each dose** (lying & standing), **rise slowly (postural hypotension)**, **don't stop abruptly** (clonidine, β-blockers), salt restriction, monitor K^+ & renal function, weight.

4.2 Antianginal Drugs

Drug	Key points
Nitroglycerin (glyceryl trinitrate, GTN)	**Venodilator (↓ preload)** + coronary dilator; **SL 0.4–0.5 mg (tablet/ spray) – onset 1–3 min**; **repeat every 5 min up to 3 doses**; if pain persists → **MI – call emergency**; **side effects – throbbing headache (most common), flushing, postural hypotension, reflex tachycardia**; **tolerance** (nitrate-free interval 8–12 hrs, e.g. remove patch at night); **store in original dark glass bottle, tightly closed, away from heat/ moisture**; tingling/ burning on tongue = potent (traditional check); **contraindicated with sildenafil/ tadalafil (PDE-5) – severe hypotension**; IV infusion – glass bottle/ non-PVC sets
Isosorbide dinitrate/ mononitrate	Oral prophylaxis
β-blockers	Reduce O_2 demand – stable angina, post-MI
CCBs	**Prinzmetal (vasospastic) angina – DOC**
Nicorandil, ranolazine, ivabradine, trimetazidine	Adjuncts
Aspirin + statin	Secondary prevention

4.3 Heart Failure & Digoxin

- **HF drugs** – **ACEI/ ARB/ ARNI (sacubitril-valsartan)**, **β-blockers (carvedilol, metoprolol succinate, bisoprolol)**, **MRA (spironolactone)**, **SGLT2 inhibitors (dapagliflozin, empagliflozin)** – ↓ mortality; **loop diuretics (symptom relief – oedema)**, digoxin (symptoms/ AF rate), ivabradine, inotropes (dobutamine) in acute.
- **Digoxin (cardiac glycoside; Digitalis lanata)**:
- **Inhibits Na^+/K^+-ATPase → ↑ intracellular Ca^{2+} → ↑ contractility (positive inotrope)**; **↑ vagal tone → ↓ HR & AV conduction (negative chronotrope/ dromotrope)**.
- Uses – **heart failure with AF (rate control)**, AF/ atrial flutter.
- **Narrow therapeutic index** – **therapeutic level 0.5–2 ng/ml** (HF: 0.5–0.9); long t½ (~36–40 hrs) – **loading (digitalisation) dose** then maintenance (0.125–0.25 mg/day).
- **Toxicity** – **GI: anorexia, nausea, vomiting (earliest)**, **visual: blurred, yellow-green vision (xanthopsia), halos**, confusion, fatigue, **arrhythmias – bradycardia, heart block, ventricular bigeminy (most common arrhythmia), VT**.
- **Factors ↑ toxicity** – **hypokalaemia (most important – diuretics)**, hypomagnesaemia, **hypercalcaemia**, renal impairment, elderly, hypothyroidism, **drugs – amiodarone, verapamil, quinidine**.
- **Antidote – digoxin-specific antibody Fab fragments (Digibind/ DigiFab)**; correct K^+, stop drug, atropine for bradycardia.
- **Nursing** – **check apical pulse for 1 full minute before giving – withhold if < 60/min (adult)** (< 90–110 infant, < 70 child) & report; check K^+, digoxin level; same time daily; teach toxicity signs & pulse counting; avoid antacids (↓ absorption).

4.4 Antiarrhythmics

Class (Vaughan Williams)	Drugs	Points
I – Na^+ channel blockers	**Lignocaine (IB – ventricular arrhythmias)**, procainamide, flecainide	Lignocaine toxicity – CNS (seizures)
II – β-blockers	Propranolol, esmolol, metoprolol	Rate control
III – K^+ channel blockers	**Amiodarone**, sotalol	**Amiodarone – broad spectrum (VF/ VT, AF)**; side effects – **pulmonary fibrosis**, **thyroid dysfunction (hypo/ hyper – iodine)**, **corneal microdeposits**, **photosensitivity, blue-grey skin**, hepatotoxicity, ↑ QT, **↑ digoxin & warfarin levels**; very long t½
IV – Ca^{2+} channel blockers	**Verapamil, diltiazem**	SVT, AF rate; not in WPW/ VT
Others	**Adenosine** – **PSVT DOC**: **6 mg rapid IV push followed by rapid 20 ml saline flush (proximal vein), then 12 mg if needed**; **t½ < 10 sec**; transient asystole, flushing, chest discomfort, dyspnoea; **contraindicated in asthma**; **atropine** (bradycardia), **digoxin**, **$MgSO_4$ (torsades de pointes)**	

4.5 Diuretics

Class	Drugs	Site	Key points
Loop ("high ceiling")	**Frusemide (furosemide)**, torsemide, bumetanide	**Thick ascending loop of Henle (Na-K-2Cl)**	**Most potent**; **acute pulmonary oedema (IV – also venodilation)**, HF, oedema, hypercalcaemia; **hypokalaemia (most common)**, hyponatraemia, hypovolaemia, hypotension, **ototoxicity (rapid IV – give slowly ≤ 4 mg/min)**, hyperuricaemia, hyperglycaemia, metabolic alkalosis; **sulfa allergy**
Thiazides	**Hydrochlorothiazide**, chlorthalidone, metolazone	Distal convoluted tubule	HTN first-line; **↓ calcium excretion (hypercalcaemia) – used in calcium kidney stones**, nephrogenic DI; hypokalaemia, hyperglycaemia, hyperuricaemia
K^+-sparing	**Spironolactone, eplerenone** (aldosterone antagonists), **amiloride, triamterene**	Collecting duct	**Hyperkalaemia** (avoid K supplements, ACEI caution), **gynaecomastia (spironolactone)**; **ascites in cirrhosis (DOC – spironolactone)**, HF, **Conn's syndrome**, hirsutism/ PCOS

Class	Drugs	Site	Key points
Osmotic	**Mannitol (20 %)**	Proximal tubule/ loop	**↓ ICP (cerebral oedema)**, ↓ IOP (acute glaucoma); **contraindicated in HF/ pulmonary oedema, anuria**; **crystals – warm & use filter**
Carbonic anhydrase inhibitors	**Acetazolamide**	PCT	**Glaucoma**, **acute mountain sickness**, epilepsy; metabolic acidosis, paraesthesia

- **Nursing** – **give in the morning** (avoid nocturia), **daily weight (best indicator of fluid loss)**, I/O, BP (postural), **serum K^+**, **K^+-rich foods (banana, orange, potato, coconut water) with loop/ thiazides** – **avoid with K-sparing**, watch digoxin toxicity with hypokalaemia.

4.6 Anticoagulants, Antiplatelets & Thrombolytics

Drug	Mechanism/ route	Monitoring	Antidote	Points
Unfractionated heparin (UFH)	Activates **antithrombin III** (↓ thrombin, Xa); **IV/ SC (never IM)**; immediate onset	**aPTT (1.5–2.5 × control)**	**Protamine sulphate (1 mg per 100 units)**	**Safe in pregnancy** (doesn't cross placenta); **HIT (heparin-induced thrombocytopenia)** – monitor platelets; SC in abdomen ≥ 5 cm from umbilicus, **don't aspirate/ massage**
LMWH – enoxaparin, dalteparin	Anti-Xa; **SC once/ twice daily**	Usually none (anti-Xa if needed)	Protamine (partial)	DVT prophylaxis/ treatment, ACS; **don't expel air bubble** from prefilled syringe; less HIT
Warfarin	**Vitamin K antagonist** (↓ factors **II, VII, IX, X**); **oral**; onset 3–5 days (overlap with heparin)	**PT/ INR – target 2–3** (**2.5–3.5 mechanical valves**)	**Vitamin K (phytonadione)**; FFP/ **PCC** for bleeding	**Teratogenic (fetal warfarin syndrome)**; **many interactions** – ↑ effect: aspirin, NSAIDs, amiodarone, metronidazole, cotrimoxazole, fluconazole, erythromycin; ↓ effect: rifampicin, carbamazepine, vitamin K; **consistent intake of green leafy vegetables (vitamin K)**; avoid alcohol; soft toothbrush, electric razor; report bleeding (gums, urine, black stools)
DOACs – dabigatran (thrombin inhibitor), **rivaroxaban, apixaban** (Xa inhibitors)	Oral, fixed dose	No routine	**Dabigatran – idarucizumab**; Xa inhibitors – andexanet alfa	AF, DVT/ PE

Antiplatelet	Points
Aspirin (low-dose 75–150 mg)	**Irreversible COX-1 inhibition → ↓ thromboxane A2** (effect lasts platelet life **7–10 days**); MI (**chew 300–325 mg non-enteric**), stroke prevention; GI bleed, **Reye's syndrome in children**, asthma (aspirin-sensitive)
Clopidogrel, prasugrel, ticagrelor	**P2Y12 (ADP) inhibitors**; with aspirin = **DAPT after stents**
Abciximab, tirofiban, eptifibatide	GP IIb/IIIa inhibitors (IV)
Dipyridamole, cilostazol	–

- **Stop aspirin/ clopidogrel ~5–7 days, warfarin ~5 days before elective surgery** (as per surgeon).

Thrombolytics (fibrinolytics)	Points
Streptokinase (antigenic – allergy, can't repeat within 1 yr), **alteplase (tPA)**, **tenecteplase (single IV bolus)**, reteplase, urokinase	**Convert plasminogen → plasmin → dissolves fibrin clot**; **STEMI – within 12 hrs (best within 3 hrs; door-to-needle ≤ 30 min)**; **acute ischaemic stroke – alteplase within 4.5 hrs** (after CT excludes haemorrhage); massive PE; **bleeding – main risk**; contraindications – active bleeding, recent surgery/ trauma, **history of haemorrhagic stroke**, severe uncontrolled HTN, aortic dissection; **antidote – tranexamic acid / aminocaproic acid** (antifibrinolytics)

- **Tranexamic acid** – antifibrinolytic – PPH (1 g IV within 3 hrs), trauma, menorrhagia, dental bleeding in haemophilia.

4.7 Hypolipidaemic Drugs

Drug	Points
Statins ("-statin") – atorvastatin, rosuvastatin, simvastatin	**HMG-CoA reductase inhibitors** – ↓ LDL most; **give at night (short-acting – simvastatin)**; **myopathy/ rhabdomyolysis (report muscle pain – check CPK)**, ↑ liver enzymes; **contraindicated in pregnancy**
Fibrates (fenofibrate, gemfibrozil)	↓ triglycerides; myopathy with statins
Ezetimibe	↓ cholesterol absorption
Niacin	Flushing (give aspirin before)
Bile acid sequestrants (cholestyramine)	Constipation; bind other drugs
PCSK9 inhibitors	Injectable

4.8 Haematinics, IV Fluids, Electrolytes & Blood

Haematinics

Drug	Points
Oral iron – ferrous sulphate (200 mg = 60 mg elemental), fumarate, gluconate	**Black stools (harmless)**, constipation/ diarrhoea, epigastric pain; **best absorbed on empty stomach** (give after food if GI upset); **with vitamin C (orange juice) ↑ absorption**; **avoid tea, coffee, milk, antacids, calcium, tetracyclines** (↓ absorption); **liquid iron – through straw (teeth staining)**; continue **3 months after Hb normal** to replenish stores; **iron poisoning in children – desferrioxamine**
Parenteral iron – iron sucrose (IV), ferric carboxymaltose (IV, up to 1 g), iron dextran (IM Z-track/ IV – test dose, anaphylaxis risk)	Malabsorption, intolerance, severe anaemia late pregnancy, CKD
Folic acid	5 mg/day (megaloblastic); 400 µg (NTD prevention)
Vitamin B12 (cyanocobalamin/ hydroxocobalamin) IM	**Pernicious anaemia – lifelong injections** (intrinsic factor absent)
Erythropoietin	**CKD anaemia**; HTN, thrombosis

IV fluids

Fluid	Type	Use
0.9 % NaCl (normal saline)	**Isotonic (308 mOsm/L)**	Shock, resuscitation, **with blood transfusion (only fluid compatible)**
Ringer lactate (Hartmann's)	Isotonic (273 mOsm/L)	**Burns, trauma, surgery, diarrhoea (closest to plasma)**; avoid in severe liver failure, hyperkalaemia caution
5 % dextrose (D5W)	Isotonic in bag (252) → **physiologically hypotonic** (free water)	Maintenance, **hypernatraemia**; **avoid in raised ICP/ head injury**
0.45 % NaCl (half NS)	**Hypotonic**	Cellular dehydration, hypernatraemia
3 % NaCl	**Hypertonic**	Severe symptomatic hyponatraemia, raised ICP (slow, central line, monitor Na – **correct ≤ 8–10 mEq/L/day – osmotic demyelination**)
10 % / 25 % / 50 % dextrose	Hypertonic	**Hypoglycaemia**
DNS (5 % dextrose in 0.9 % saline)	Hypertonic	Maintenance
Colloids – albumin, dextran, gelatin (Haemaccel), HES	–	Volume expansion; **dextran – anaphylaxis, bleeding**

- **Potassium chloride** – **never IV push**; dilute; **max 10 mEq/hr (peripheral)**; **K^+ ≤ 40 mEq/L** peripheral; ensure urine output (≥ 30 ml/hr); cardiac monitor for high rates; **oral KCl – dilute in water/ juice, after food (GI irritation)**. Normal serum K^+ **3.5–5.0 mEq/L**.

- **10 % calcium gluconate** – **slow IV (over 10 min)** – hypocalcaemia, **hyperkalaemia (cardioprotection)**, **$MgSO_4$ toxicity antidote**; **don't mix with bicarbonate** (precipitates); **calcium chloride – central line (irritant)**.
- **Sodium bicarbonate (7.5/ 8.4 %)** – severe metabolic acidosis, hyperkalaemia, TCA overdose.
- **Hyperkalaemia treatment** – **calcium gluconate (stabilises myocardium – first)**, **insulin + dextrose**, salbutamol nebulisation, bicarbonate (shift into cells), **resins/ diuretics/ dialysis (remove)**.

Blood transfusion – nursing points

- **Consent, blood grouping & cross-matching**, **verify at bedside by 2 staff** (patient ID, blood group, bag number, expiry).
- **Start within 30 min of issue; complete 1 unit within 4 hrs**; **18–20 G cannula**; **blood set with filter (170–200 µm)**; **only normal saline** (no dextrose – haemolysis; no RL – clotting with calcium; no drugs in line).
- **Vitals before, 15 min after start, then hourly & at end**; **first 15 min slowly (most reactions)** – stay with patient.
- **Reactions** – **febrile non-haemolytic (most common)**, **allergic (urticaria – antihistamine)**, **acute haemolytic (ABO incompatibility – most dangerous: fever, chills, back/ loin pain, chest tightness, hypotension, haemoglobinuria, DIC)**, **TACO (circulatory overload)**, **TRALI**, anaphylaxis (IgA-deficient), bacterial contamination, delayed haemolytic, **hyperkalaemia, hypocalcaemia (citrate) in massive transfusion**, infection (HIV, HBV, HCV, malaria, syphilis).
- **If reaction – STOP transfusion immediately**, **keep IV line open with NS (new tubing)**, inform doctor & blood bank, vitals, **send bag + set + patient blood & urine samples**, document.
- **Storage** – **packed RBCs/ whole blood 2–6 °C (CPDA-1: 35 days; SAGM: 42 days)**; **platelets 20–24 °C with continuous agitation – 5 days**; **FFP ≤ −18 °C (ideally −30 °C) – 1 yr; thaw at 37 °C, use within 6 hrs**; **cryoprecipitate** – fibrinogen, factor VIII.
- **1 unit packed RBC ↑ Hb by ~1 g/dl** (adult).

PYQ style

- ACE inhibitor common side effect – **dry cough**; serious – **angioedema**.
- ACEI/ ARB in pregnancy – **contraindicated**.
- Amlodipine side effect – **ankle oedema**.
- Hypertensive emergency – **IV labetalol/ nitroprusside**; nitroprusside – **protect from light, cyanide toxicity**.
- Nitroglycerin SL – **every 5 min, max 3 doses**; side effect – **headache**; stored in – **dark glass bottle**.
- Nitrates contraindicated with – **sildenafil**.
- Prinzmetal angina – **CCB**.
- Digoxin mechanism – **Na^+/K^+-ATPase inhibition**; level – **0.5–2 ng/ml**.
- Earliest digoxin toxicity – **anorexia, nausea, vomiting**; visual – **yellow-green halos**.
- Digoxin toxicity ↑ with – **hypokalaemia**; antidote – **Digibind (Fab)**.
- Withhold digoxin – **apical pulse < 60/min**.
- PSVT DOC – **adenosine 6 mg rapid IV push + flush**.
- Amiodarone – **pulmonary fibrosis, thyroid dysfunction**.
- Most potent diuretic – **frusemide (loop)**; side effect – **hypokalaemia, ototoxicity**.
- K^+-sparing – **spironolactone**; side effect – **hyperkalaemia, gynaecomastia**.
- Raised ICP – **mannitol**; mountain sickness – **acetazolamide**.
- Diuretic timing – **morning**; best fluid status indicator – **daily weight**.
- Heparin monitoring – **aPTT**; antidote – **protamine sulphate**.
- Warfarin monitoring – **INR (2–3)**; antidote – **vitamin K**.
- Anticoagulant safe in pregnancy – **heparin**; teratogenic – **warfarin**.
- Aspirin mechanism – **irreversible COX inhibition**; children – **Reye's**.
- Thrombolytic in stroke – **alteplase within 4.5 hrs**; MI – **within 12 hrs**.
- Thrombolytic antidote – **tranexamic acid**.
- Statin side effect – **myopathy/ rhabdomyolysis**; timing – **night**.
- Iron + vitamin C – **↑ absorption**; iron stool – **black**; liquid iron – **straw**.
- Pernicious anaemia – **lifelong IM B12**.
- Only fluid with blood – **normal saline**; complete unit in – **4 hrs**.
- Most common transfusion reaction – **febrile non-haemolytic**; most dangerous – **acute haemolytic (ABO)**.
- First action in reaction – **stop transfusion, keep line open with NS**.
- Platelets storage – **20–24 °C with agitation**; RBCs – **2–6 °C**.
- KCl – **never IV push; max 10 mEq/hr**.
- Hyperkalaemia first drug – **calcium gluconate**.

5 Respiratory & Gastrointestinal Drugs

5.1 Drugs for Asthma & COPD

Class	Drugs	Points
SABA (short-acting β2 agonist) – reliever	**Salbutamol**, terbutaline, levalbuterol	**Acute attack – DOC**; inhaled/ nebulised; tremor, tachycardia, palpitation, **hypokalaemia**
LABA	Salmeterol, formoterol, vilanterol	Maintenance – **always with ICS** in asthma
Anticholinergics	**Ipratropium (SAMA)**, **tiotropium (LAMA – COPD)**	**COPD – bronchodilator of choice**; dry mouth
Inhaled corticosteroids (ICS) – controller	**Budesonide, beclomethasone, fluticasone**	**Most effective long-term controller in asthma**; **oral thrush (candidiasis), hoarseness – rinse mouth & gargle after use**, use spacer
Systemic steroids	**Prednisolone (oral), hydrocortisone, methylprednisolone (IV)**	Acute severe exacerbation
Methylxanthines	**Theophylline (oral), aminophylline (IV)**	**Narrow TI – therapeutic 10–20 µg/ml** (5–15 now preferred); **toxicity – nausea, vomiting, tachycardia, arrhythmias, seizures**; **aminophylline IV slowly (≥ 20 min)**; **↑ levels – ciprofloxacin, erythromycin, cimetidine; ↓ levels – smoking, rifampicin**; avoid caffeine
Leukotriene antagonists	**Montelukast** (bedtime), zafirlukast	Aspirin-induced, exercise-induced asthma, allergic rhinitis; neuropsychiatric effects
Mast cell stabilisers	**Sodium cromoglycate**	Prophylaxis only (not acute)
Anti-IgE / biologics	Omalizumab, mepolizumab	Severe asthma
$MgSO_4$ IV	–	Acute severe asthma

- **Inhaler order – bronchodilator first, wait 5 min, then steroid.**
- **Acute severe asthma – O_2, nebulised salbutamol (+ ipratropium), systemic steroids, IV $MgSO_4$,** (aminophylline).

5.2 Cough, Cold & Antihistamines

Group	Drugs	Points
Antitussives (cough suppressants – dry cough)	**Codeine** (opioid – constipation, dependence), **dextromethorphan**, pholcodine	**Not in productive cough**; codeine avoided < 12 yrs
Expectorants	**Guaifenesin**, ammonium chloride, potassium iodide	↑ volume of secretions

Group	Drugs	Points
Mucolytics	**Bromhexine, ambroxol, N-acetylcysteine (also paracetamol antidote)**, carbocisteine	Thin sputum
Nasal decongestants	Oxymetazoline, xylometazoline (topical), pseudoephedrine (oral)	**Rebound congestion (rhinitis medicamentosa) if > 3–5 days**
H1 antihistamines – 1st gen (sedating)	**Chlorpheniramine (CPM), promethazine, diphenhydramine, hydroxyzine, cyproheptadine**	**Sedation – avoid driving**, anticholinergic (dry mouth, urinary retention), **promethazine – antiemetic, motion sickness**; cyproheptadine – appetite stimulant, serotonin syndrome
H1 – 2nd gen (non-sedating)	**Cetirizine, levocetirizine, loratadine, fexofenadine**, desloratadine	Allergic rhinitis, urticaria

5.3 Peptic Ulcer & Acid-Related Disorders

Class	Drugs	Points
Antacids	**Magnesium hydroxide (diarrhoea), aluminium hydroxide (constipation)** – combined to balance; calcium carbonate (constipation, rebound acid, milk-alkali), sodium bicarbonate (systemic alkalosis, CO_2 belching)	**Give 1–3 hrs after meals & at bedtime**; **separate other drugs by 1–2 hrs** (↓ absorption – **tetracyclines, fluoroquinolones, iron, digoxin**); Al – hypophosphataemia; Mg – avoid in renal failure
H2 blockers	**Ranitidine (withdrawn – NDMA), famotidine**, cimetidine (enzyme inhibitor, gynaecomastia)	Reduce acid
Proton pump inhibitors (PPIs, "-prazole")	**Omeprazole, pantoprazole, esomeprazole, rabeprazole, lansoprazole**	**Most potent acid suppressants**; **give 30–60 min before breakfast** (empty stomach); GERD, PUD, Zollinger-Ellison, **H. pylori**, stress ulcer prophylaxis, with NSAIDs; long-term – **↓ Mg, B12 deficiency, osteoporosis/ fractures, C. difficile, pneumonia**
Sucralfate	Coats ulcer base	**Empty stomach (1 hr before meals)**; constipation; separate other drugs
Misoprostol (PGE1)	Cytoprotective	**NSAID-induced ulcer prevention**; **contraindicated in pregnancy (abortion)**; diarrhoea
Bismuth	–	Black tongue & stools

- **H. pylori eradication (triple therapy, 14 days)** – **PPI + clarithromycin + amoxicillin** (or metronidazole); **quadruple** – PPI + bismuth + tetracycline + metronidazole.

5.4 Antiemetics

Class	Drugs	Use / side effects
5-HT3 antagonists	**Ondansetron**, granisetron, palonosetron	**Chemotherapy-induced (DOC), post-op, radiotherapy, gastroenteritis**; headache, constipation, **QT prolongation**
D2 antagonists (prokinetic)	**Metoclopramide** (crosses BBB), **domperidone** (doesn't cross BBB – fewer EPS)	GERD, gastroparesis, migraine; **metoclopramide – EPS (acute dystonia – esp. young), tardive dyskinesia**; avoid in GI obstruction
Phenothiazines	**Prochlorperazine**, promethazine, chlorpromazine	Vertigo, nausea; sedation, EPS
Antihistamines/ anticholinergics	**Promethazine, dimenhydrinate, meclizine, hyoscine**	**Motion sickness** (take 30–60 min before travel)
NK1 antagonists	**Aprepitant**	Chemo (delayed emesis)
Corticosteroids	Dexamethasone	Chemo (adjunct)
Doxylamine + pyridoxine	–	**Nausea & vomiting of pregnancy**
Cannabinoids	Dronabinol	Chemo

5.5 Laxatives (Purgatives)

Type	Drugs	Points
Bulk-forming	**Isabgol (psyllium/ ispaghula)**, methylcellulose, bran	**Safest, most physiological**; **take with plenty of water (≥ 1 glass)** – else obstruction; onset 1–3 days
Stool softeners (emollients)	**Docusate sodium**, liquid paraffin (lubricant)	**Liquid paraffin – lipoid pneumonia (aspiration – avoid at bedtime/ in elderly, children), ↓ fat-soluble vitamins (A, D, E, K) absorption, anal seepage**
Osmotic	**Lactulose** (**hepatic encephalopathy – ↓ ammonia** – titrate to 2–3 soft stools/day), **magnesium sulphate/ hydroxide (milk of magnesia)**, **polyethylene glycol (PEG – bowel prep)**, sodium phosphate	Dehydration, electrolyte imbalance
Stimulant	**Bisacodyl, senna, sodium picosulfate**, castor oil	Cramping; chronic use → **cathartic colon**; **bisacodyl – don't chew, avoid with antacids/ milk**
Others	Lubiprostone, prucalopride	–

- **Contraindications** – **acute abdomen/ suspected appendicitis, intestinal obstruction**, undiagnosed abdominal pain; **laxative abuse** (eating disorders) → hypokalaemia.
- **Enemas** – **soap water, saline, phosphate (Fleet)**, glycerine suppositories.

5.6 Antidiarrhoeals & Others

- **ORS + zinc (mainstay)**, **probiotics**, **racecadotril** (antisecretory – children ok).
- **Loperamide** (opioid – ↓ motility; **not in children < 2–4 yrs, dysentery/ bloody diarrhoea, fever** – toxic megacolon), **diphenoxylate + atropine (Lomotil)**.
- **Antispasmodics** – **dicyclomine, hyoscine butylbromide, drotaverine, mebeverine** (IBS).
- **IBD** – **5-ASA (mesalamine, sulfasalazine – orange urine, oligospermia, folate ↓)**, steroids, azathioprine, biologics (infliximab).
- **Liver** – **lactulose & rifaximin (hepatic encephalopathy)**, ursodeoxycholic acid (gallstones, cholestasis), **propranolol (varices)**, **octreotide/ terlipressin (variceal bleed)**, **vitamin K**.
- **Pancreatic enzymes (pancreatin)** – with meals (cystic fibrosis, chronic pancreatitis).
- **Anorectal** – stool softeners, sitz bath.

PYQ style

- Asthma acute attack – **salbutamol**; long-term controller – **inhaled corticosteroid**.
- ICS side effect – **oral thrush → rinse mouth**.
- Inhaler sequence – **bronchodilator before steroid**.
- COPD bronchodilator – **tiotropium/ ipratropium**.
- Theophylline therapeutic level – **10–20 µg/ml**; aminophylline – **slow IV**.
- Montelukast – **leukotriene antagonist (bedtime)**.
- Cromolyn – **prophylaxis only**.
- Dry cough suppressant – **codeine/ dextromethorphan**.
- Mucolytic & paracetamol antidote – **N-acetylcysteine**.
- Non-sedating antihistamine – **cetirizine/ loratadine/ fexofenadine**.
- Antacid causing diarrhoea – **magnesium**; constipation – **aluminium**.
- Antacids reduce absorption of – **tetracyclines, iron, fluoroquinolones**.
- PPI – **before breakfast**; most potent acid suppressant – **PPI**.
- H. pylori – **PPI + clarithromycin + amoxicillin (14 days)**.
- NSAID ulcer prevention – **misoprostol (contraindicated pregnancy)**.
- Chemotherapy vomiting DOC – **ondansetron**.
- Metoclopramide side effect – **acute dystonia/ EPS**.
- Morning sickness – **doxylamine + pyridoxine**.
- Safest laxative – **bulk-forming (isabgol) with plenty of water**.
- Liquid paraffin – **lipoid pneumonia, ↓ fat-soluble vitamins**.
- Hepatic encephalopathy – **lactulose**.
- Laxative contraindicated in – **acute abdomen/ obstruction**.
- Loperamide avoided in – **young children, dysentery**.
- Sulfasalazine – **orange urine**.

6 CNS Drugs – Analgesics, Antiepileptics, Anaesthetics & Others

6.1 Opioid Analgesics

- **Morphine** (prototype – from **opium poppy**; μ-receptor agonist).
- Uses – **severe pain (MI, cancer, post-op, trauma)**, **acute pulmonary oedema (LVF)**, palliative dyspnoea, pre-anaesthetic.
- **Effects/ side effects** – analgesia, euphoria, sedation, **respiratory depression (most dangerous – check RR before giving; withhold if RR < 12/min)**, **pin-point pupils (miosis)**, **constipation (most common – no tolerance develops)**, **nausea, vomiting**, **urinary retention**, **pruritus (histamine release)**, hypotension, bradycardia, ↑ ICP (due to CO_2 retention), cough suppression, **tolerance & dependence**.
- **Contraindications/ caution** – **head injury (masks signs, ↑ ICP)**, **bronchial asthma/ COPD**, **acute abdomen (masks diagnosis)**, hypothyroidism, **biliary colic (spasm of sphincter of Oddi – pethidine preferred)**, liver disease, infants/ elderly, pregnancy near term (neonatal depression).
- **Overdose triad – coma, respiratory depression, pin-point pupils → naloxone 0.4–2 mg IV** (short-acting – repeat/ infusion).
- **Other opioids** – **pethidine (meperidine)** (labour analgesia, **neonatal respiratory depression if within 2–4 hrs of delivery**, normeperidine seizures; avoid with MAOI), **fentanyl** (100 × morphine; **transdermal patch** – chronic pain; IV anaesthesia), **tramadol** (weak opioid + SNRI – **seizures, serotonin syndrome**), **codeine** (cough, mild pain), **buprenorphine** (partial agonist – OST, SL), **methadone** (OST), **pentazocine** (dysphoria), oxycodone, tapentadol.
- **Naloxone** (antagonist – overdose, **neonatal opioid depression 0.1 mg/kg**), **naltrexone** (oral – alcohol/ opioid dependence).
- **WHO analgesic ladder (cancer pain)** – **Step 1: non-opioid (paracetamol/ NSAID) ± adjuvant → Step 2: weak opioid (codeine, tramadol)** + non-opioid → **Step 3: strong opioid (morphine, fentanyl)** + non-opioid; **"by mouth, by the clock (regular), by the ladder, for the individual, attention to detail"**; adjuvants – antidepressants, anticonvulsants, steroids; **prophylactic laxative with opioids**.
- **PCA (patient-controlled analgesia)** – lockout interval prevents overdose; **only patient presses the button**.

6.2 Non-Opioid Analgesics

NSAIDs (non-steroidal anti-inflammatory drugs)

- **Mechanism – inhibit cyclo-oxygenase (COX) → ↓ prostaglandins → analgesic, antipyretic, anti-inflammatory** (+ antiplatelet – aspirin).
- **Drugs** – **aspirin (acetylsalicylic acid – irreversible COX)**, **ibuprofen, diclofenac, naproxen, ketorolac, indomethacin, mefenamic acid (dysmenorrhoea)**, piroxicam; **selective COX-2 – celecoxib, etoricoxib** (less GI, **↑ CV risk – MI, stroke**).
- **Side effects** – **GI – gastritis, peptic ulcer, GI bleeding (most common) → give after food/ with PPI**; **renal (↓ GFR, Na & water retention, AKI – esp. elderly, dehydration)**; **↑ BP, ↓ efficacy of**

antihypertensives/ diuretics; **bleeding**; **bronchospasm (aspirin-sensitive asthma)**; hepatotoxicity; **premature closure of ductus arteriosus (3rd trimester – avoid)**.

- **Aspirin special** – **Reye's syndrome (children < 16 yrs with viral illness – avoid)**; **salicylism (high dose) – tinnitus, deafness, vertigo, nausea**; **aspirin poisoning – respiratory alkalosis → metabolic acidosis, hyperthermia** – Rx **alkaline diuresis (sodium bicarbonate)**, haemodialysis; **low dose (75–150 mg) – antiplatelet**; stop 7 days before surgery.
- **Indomethacin** – PDA closure in preterm, gout; **ketorolac** – potent, short-term (≤ 5 days).

Paracetamol (acetaminophen)

- **Analgesic & antipyretic, weak anti-inflammatory**; **safe in children, pregnancy, peptic ulcer, bleeding disorders, asthma** – **antipyretic of choice in children & dengue**.
- Dose – **adult 500–1000 mg every 4–6 hrs, max 4 g/day** (≤ 2–3 g in alcoholics/ liver disease); **child 10–15 mg/kg/dose**.
- **Toxicity (overdose > 150 mg/kg or > 7.5–10 g adult)** – toxic metabolite **NAPQI** depletes **glutathione → hepatic necrosis** (24–72 hrs); early asymptomatic/ nausea → **↑ liver enzymes, jaundice, liver failure (day 3–4)**; **Rumack-Matthew nomogram** (level at 4 hrs).
- **Antidote – N-acetylcysteine (NAC)** – most effective **within 8–10 hrs** (useful up to 24 hrs+); oral/ IV; **activated charcoal within 1 hr**.

6.3 Antiepileptics

Drug	Use	Key side effects / nursing
Phenytoin	GTCS, focal, status epilepticus (after BZD)	**Gingival hyperplasia (oral hygiene)**, **hirsutism**, coarse facies, **nystagmus, ataxia, diplopia (toxicity)**, megaloblastic anaemia (folate), osteomalacia, **SJS**, teratogenic (fetal hydantoin syndrome); **therapeutic 10–20 µg/ml; zero-order kinetics**; **IV only in NORMAL SALINE (precipitates in dextrose), ≤ 50 mg/min (adult), flush before/ after, cardiac monitoring (hypotension, arrhythmia)**; **purple glove syndrome**; enzyme inducer; **never IM**
Fosphenytoin	Prodrug – IV/ IM	Safer
Carbamazepine	**Focal seizures (DOC)**, **trigeminal neuralgia (DOC)**, bipolar	**Agranulocytosis/ aplastic anaemia**, hyponatraemia (SIADH), **SJS (HLA-B*1502 – Asians)**, diplopia, enzyme inducer (OCP failure)
Sodium valproate	**Broad spectrum – GTCS, absence, myoclonic**, bipolar	**Hepatotoxicity (children < 2 yrs)**, **pancreatitis**, weight gain, **hair loss**, tremor, thrombocytopenia, **NTD (most teratogenic – avoid in pregnancy)**, PCOS; enzyme inhibitor
Ethosuximide	**Absence seizures (petit mal) DOC**	GI, drowsiness
Phenobarbitone	**Neonatal seizures (DOC)**, status	**Sedation**, hyperactivity in children, dependence, enzyme inducer
Lamotrigine	Broad, bipolar depression	**SJS rash – slow titration**

Drug	Use	Key side effects / nursing
Levetiracetam	Broad – safe, few interactions; pregnancy	Behavioural changes, irritability
Topiramate	Broad, migraine	Weight loss, kidney stones, cognitive slowing
Clonazepam, clobazam	Myoclonic, adjunct	Sedation
Gabapentin, pregabalin	Neuropathic pain, focal	Dizziness
Vigabatrin / ACTH	**Infantile spasms**	Visual field defects (vigabatrin)
Magnesium sulphate	**Eclampsia (DOC)**	Knee jerk, RR, urine output

- **Status epilepticus** – **IV lorazepam 4 mg (0.1 mg/kg) / diazepam 10 mg / IM/ buccal midazolam** → **phenytoin/ fosphenytoin 20 mg/kg, valproate, levetiracetam** → phenobarbitone → anaesthesia (midazolam/ propofol/ thiopentone infusion).
- **Teaching** – **never stop abruptly (status)**, regular timing, **drug levels**, avoid alcohol, **don't drive** until seizure-free (as per law), medical ID, **contraception (enzyme inducers ↓ OCP effect)**, **folic acid 5 mg before & during pregnancy**, report rash/ sore throat/ bleeding.

6.4 Antiparkinson Drugs

- **Parkinson's** – **↓ dopamine (substantia nigra)**, relative ↑ ACh.

Drug	Points
Levodopa + carbidopa (Syndopa)	**Most effective**; levodopa – dopamine precursor (crosses BBB); **carbidopa – peripheral decarboxylase inhibitor (↓ dose, ↓ nausea)**; side effects – **nausea, postural hypotension, dyskinesias, "on-off" phenomenon, "wearing off", hallucinations, dark urine/ sweat**; **avoid high-protein meals with dose** (↓ absorption); **pyridoxine (B6) ↓ effect of levodopa alone** (not with carbidopa); **MAOI-A → hypertensive crisis**; don't stop abruptly (NMS-like)
Dopamine agonists	**Pramipexole, ropinirole, bromocriptine** – impulse control disorders, sleep attacks
MAO-B inhibitors	**Selegiline, rasagiline**
COMT inhibitors	Entacapone
Anticholinergics	**Trihexyphenidyl, benztropine** – tremor, **drug-induced parkinsonism**; avoid elderly (confusion)
Amantadine	Dyskinesia; also antiviral

6.5 Sedative-Hypnotics & Psychotropics (summary)

- **Benzodiazepines** – **diazepam, lorazepam, alprazolam, midazolam (IV sedation, short-acting), chlordiazepoxide, clonazepam, nitrazepam** – anxiety, insomnia, seizures, alcohol withdrawal, pre-

medication, muscle spasm; **sedation, respiratory depression (with alcohol/ opioids), dependence, paradoxical agitation (elderly), falls** – **antidote flumazenil**.

- **Z-drugs (zolpidem, zopiclone)** – insomnia; **barbiturates** – largely obsolete (phenobarbitone, thiopentone).
- **Antipsychotics, antidepressants, lithium** – see Mental Health notes (EPS, NMS, lithium 0.6–1.2 mEq/L, MAOI cheese reaction).

6.6 Anaesthetics

General anaesthesia (GA)

- **Stages of ether anaesthesia (Guedel)** – **I analgesia**, **II excitement/ delirium (dangerous)**, **III surgical anaesthesia (4 planes)**, **IV medullary paralysis (respiratory arrest)**.
- **Components** – unconsciousness, analgesia, amnesia, muscle relaxation, loss of reflexes.

Agent	Points
Thiopentone sodium (IV barbiturate)	**Rapid induction (ultra-short – redistribution)**; laryngospasm, hypotension; **intra-arterial injection → gangrene**; extravasation – necrosis; contraindicated – porphyria
Propofol	**Most common IV induction**; milky white (**soya/ egg lecithin**); rapid smooth recovery, antiemetic; **pain on injection, hypotension**; **propofol infusion syndrome**; ICU sedation
Ketamine	**Dissociative anaesthesia** (eyes open, analgesia); **↑ BP & HR (good in shock/ hypovolaemia)**, **bronchodilator (asthma)**; **emergence delirium/ hallucinations (give BZD)**, ↑ ICP, ↑ secretions
Etomidate	Cardiovascular stable; adrenal suppression
Midazolam	Sedation, amnesia
Nitrous oxide (N_2O – "laughing gas")	Weak anaesthetic, good analgesic; **Entonox (50 % N_2O + 50 % O_2) – labour**; **diffusion hypoxia** at end (give 100 % O_2); B12 inactivation
Halothane	**Halothane hepatitis**, arrhythmias (sensitises heart to adrenaline), uterine relaxation
Isoflurane, sevoflurane, desflurane	**Sevoflurane – inhalational induction in children (pleasant, non-irritant)**; desflurane – rapid recovery, airway irritation
Ether (historical)	Explosive, irritant – **first public demonstration – William Morton, 16 Oct 1846 (Ether Day)**

- **Pre-anaesthetic medication** – **anxiolysis (BZD)**, **↓ secretions (atropine/ glycopyrrolate)**, **analgesia (opioids)**, **antiemetic (ondansetron)**, **↓ gastric acid (H2 blocker/ PPI, metoclopramide)** – prevents **Mendelson's syndrome (aspiration)**.

Neuromuscular blockers (muscle relaxants)

Type	Drugs	Points
Depolarising	**Succinylcholine (suxamethonium)**	**Rapid onset (~1 min), short duration (5–10 min) – rapid sequence intubation, ECT**; **fasciculations, post-op myalgia, hyperkalaemia (burns, crush, paraplegia – avoid)**, **malignant hyperthermia trigger**, ↑ IOP/ ICP; **prolonged apnoea in pseudocholinesterase deficiency**; **not reversed by neostigmine**; **refrigerate**
Non-depolarising (competitive)	**Atracurium (Hofmann elimination – renal/ hepatic failure), vecuronium, rocuronium**, pancuronium, cisatracurium	**Reversed by neostigmine + glycopyrrolate/ atropine**; **sugammadex – reverses rocuronium/ vecuronium**
Centrally acting muscle relaxants	**Baclofen, tizanidine, diazepam, chlorzoxazone**	Spasticity, back spasm
Dantrolene	Direct (ryanodine receptor)	**Malignant hyperthermia (DOC)**, NMS, spasticity

- **Malignant hyperthermia** – **genetic (autosomal dominant – RYR1)**, triggered by **volatile anaesthetics (halothane) + succinylcholine** → **↑ $EtCO_2$ (earliest sign), masseter spasm, rigidity, tachycardia, hyperthermia (late), hyperkalaemia, acidosis, rhabdomyolysis** → **stop trigger, 100 % O_2, IV dantrolene 2.5 mg/kg**, cooling, treat hyperkalaemia.

Local anaesthetics

Drug	Points
Lignocaine (lidocaine)	**Most widely used**; surface (2–4 %), infiltration (**1 %**), nerve block, **spinal (5 % heavy)**, antiarrhythmic (IB); **max dose 3 mg/kg (4.5 mg/kg) plain, 7 mg/kg with adrenaline**
Bupivacaine	**Long-acting**; spinal (0.5 % heavy), epidural; **most cardiotoxic – never IV**
Ropivacaine	Less cardiotoxic
Prilocaine	**Methaemoglobinaemia** (Rx methylene blue)
EMLA cream (lidocaine + prilocaine)	**Topical before venipuncture (apply 60 min, occlusive dressing)**
Cocaine	Topical (nasal) – vasoconstrictor; abuse
Benzocaine	Topical

- **Adrenaline added** – ↓ systemic absorption, prolongs action, ↓ bleeding; **avoid in end-arteries – fingers, toes, ear, nose, penis (gangrene)**.
- **LA systemic toxicity (LAST)** – **circumoral numbness, tingling, metallic taste, tinnitus, slurred speech → seizures → CNS depression, arrhythmia, cardiac arrest** → **stop injection, airway, BZD for seizures, 20 % lipid emulsion (Intralipid)**.
- **Spinal anaesthesia** – complications – **hypotension (most common – preload fluids, vasopressor)**, **post-dural puncture headache (lie flat, hydration, caffeine, epidural blood patch)**, urinary retention, high spinal; **site L3–L4/ L4–L5**.

6.7 Other CNS Drugs

- **Migraine** – **acute: paracetamol/ NSAIDs, triptans (sumatriptan – contraindicated in CAD)**, ergotamine (vasospasm); **prophylaxis: propranolol, flunarizine, amitriptyline, topiramate, valproate**.
- **Myasthenia** – pyridostigmine.
- **Vertigo** – betahistine, cinnarizine, prochlorperazine.
- **Dementia** – donepezil, memantine.
- **ADHD** – methylphenidate; **narcolepsy** – modafinil.
- **CNS stimulants** – caffeine (apnoea of prematurity – caffeine citrate), amphetamines.

PYQ style

- Most dangerous side effect of morphine – **respiratory depression**; most common – **constipation**.
- Withhold morphine if RR – **< 12/min**.
- Opioid pupils – **pin-point**; overdose antidote – **naloxone**.
- Morphine contraindicated in – **head injury, asthma**.
- Morphine in acute pulmonary oedema – **yes (↓ preload, anxiety)**.
- Pethidine near delivery – **neonatal respiratory depression**.
- WHO pain ladder step 3 – **strong opioid (morphine)**.
- NSAID mechanism – **COX inhibition**; commonest side effect – **GI ulcer/ bleeding**.
- Aspirin in children – **Reye's syndrome**; high-dose toxicity – **tinnitus (salicylism)**.
- Selective COX-2 inhibitor – **celecoxib (↑ CV risk)**.
- Paracetamol max dose – **4 g/day**; toxicity – **liver necrosis**; antidote – **N-acetylcysteine**.
- Antipyretic in dengue – **paracetamol**.
- Phenytoin – **gingival hyperplasia**; IV only in – **normal saline**; level – **10–20 µg/ml**.
- Trigeminal neuralgia DOC – **carbamazepine**.
- Absence seizure DOC – **ethosuximide (valproate)**.
- Neonatal seizures DOC – **phenobarbitone**.
- Most teratogenic antiepileptic – **valproate (NTD)**.
- Status epilepticus first drug – **IV lorazepam/ diazepam**.
- Parkinson's most effective – **levodopa + carbidopa**; carbidopa – **peripheral decarboxylase inhibitor**.
- Levodopa effect ↓ by – **pyridoxine, high-protein meal**.
- BZD antidote – **flumazenil**.
- Dissociative anaesthetic – **ketamine (↑ BP, bronchodilator)**.
- Commonest IV induction agent – **propofol (milky, egg/ soya)**.
- Inhalational induction in children – **sevoflurane**.
- Laughing gas – **N_2O**; Entonox – **50 % N_2O + 50 % O_2**.
- Halothane – **hepatitis**.
- Depolarising relaxant – **succinylcholine**; triggers – **malignant hyperthermia**.
- Malignant hyperthermia DOC – **dantrolene**; earliest sign – **↑ $EtCO_2$**.
- Non-depolarising reversal – **neostigmine**.
- Lignocaine max – **3–4.5 mg/kg plain, 7 mg/kg with adrenaline**.
- Most cardiotoxic LA – **bupivacaine**; LA toxicity Rx – **lipid emulsion**.
- EMLA applied – **60 min before**.
- Spinal anaesthesia commonest complication – **hypotension**; headache – **lie flat, fluids**.

7 Antimicrobial Drugs

7.1 Principles of Antimicrobial Therapy

- **Bactericidal** (kill) – **penicillins, cephalosporins, aminoglycosides, fluoroquinolones, vancomycin, metronidazole, rifampicin, INH, cotrimoxazole**; **bacteriostatic** (inhibit growth) – **tetracyclines, macrolides (low dose), chloramphenicol, sulfonamides alone, clindamycin, linezolid**.
- **Mechanisms** – **cell wall synthesis inhibitors** (β-lactams, vancomycin), **protein synthesis** (aminoglycosides, tetracyclines – 30S; macrolides, chloramphenicol, clindamycin, linezolid – 50S), **DNA gyrase** (fluoroquinolones), **folate synthesis** (sulfonamides, trimethoprim), **cell membrane** (polymyxin, amphotericin), **RNA polymerase** (rifampicin).
- **Culture & sensitivity before first dose** (but don't delay in sepsis – **antibiotics within 1 hr in septic shock**); **complete the full course**; correct dose & interval (**maintain blood levels – give on time**); **test dose/ ask allergy history**; **superinfection** (C. difficile – pseudomembranous colitis, oral/ vaginal candidiasis); **antimicrobial resistance (AMR)** – **antimicrobial stewardship**, **Red Line campaign** (India – red vertical line on Rx-only antibiotics), **AWaRe classification (WHO – Access, Watch, Reserve)**.
- **Prophylaxis** – **surgical (cefazolin within 60 min before incision)**, rheumatic fever (benzathine penicillin), meningococcal contacts (rifampicin), PEP.

7.2 β-Lactam Antibiotics

Group	Drugs	Points
Natural penicillins	**Penicillin G (benzylpenicillin – IV/ IM)**, **benzathine penicillin (long-acting depot – IM deep, every 21 days for RF; syphilis 2.4 MU)**, procaine penicillin, penicillin V (oral)	**Penicillin – Alexander Fleming 1928**; **anaphylaxis (most dangerous – test dose, keep adrenaline ready, observe 30 min)**; Jarisch-Herxheimer reaction (syphilis)
Penicillinase-resistant	**Cloxacillin, flucloxacillin**, methicillin	**Staph aureus** (skin infections)
Aminopenicillins	**Amoxicillin, ampicillin**	Broad; **rash (esp. in infectious mononucleosis)**, diarrhoea; **amoxicillin + clavulanic acid (β-lactamase inhibitor) – Augmentin**
Antipseudomonal	**Piperacillin + tazobactam**	Hospital infections
Cephalosporins – 1st gen	**Cefazolin, cephalexin**	**Surgical prophylaxis**, skin
– 2nd gen	Cefuroxime, cefoxitin	RTI
– **3rd gen**	**Ceftriaxone (meningitis, typhoid, gonorrhoea; once daily; don't mix with calcium solutions – precipitates in neonates)**, **cefotaxime (neonates)**, **ceftazidime (Pseudomonas)**, **cefixime (oral – typhoid, UTI)**, cefoperazone	–
– 4th / 5th gen	**Cefepime** / ceftaroline (MRSA)	–

Group	Drugs	Points
Carbapenems	**Meropenem, imipenem (+ cilastatin – seizures)**, ertapenem	**Reserve – ESBL, MDR**
Monobactam	Aztreonam	Safe in penicillin allergy

- **Cross-allergy** penicillin ↔ cephalosporins (~1–2 %; avoid if anaphylaxis).
- **Cephalosporins + alcohol** (cefoperazone) – disulfiram-like reaction.

7.3 Other Antibacterials

Drug/ group	Key uses	Key adverse effects / nursing
Aminoglycosides – gentamicin, amikacin, streptomycin, tobramycin, neomycin (oral – bowel prep, hepatic coma)	Gram-negative sepsis, **TB (streptomycin)**, endocarditis (synergy)	**Nephrotoxicity** (monitor creatinine, urine output, hydrate) & **ototoxicity (8th nerve – vestibular & cochlear; irreversible – tinnitus, vertigo, deafness)**, **neuromuscular blockade (apnoea – with anaesthesia/ $MgSO_4$; myasthenia)**; **once-daily dosing**; **peak & trough levels**; **avoid in pregnancy (fetal deafness)**; **not absorbed orally**
Macrolides – erythromycin, azithromycin, clarithromycin	**Atypical pneumonia (Mycoplasma), pertussis, diphtheria, chlamydia, penicillin allergy**, H. pylori (clarithro), typhoid & trachoma (azithro)	**GI upset (erythromycin – motilin agonist)**, **QT prolongation**, **enzyme inhibition (↑ warfarin, theophylline, digoxin)**, cholestatic jaundice (erythromycin estolate)
Tetracyclines – doxycycline, tetracycline, minocycline	**Rickettsia (scrub typhus – DOC), cholera, chlamydia, acne, Lyme, leptospirosis, malaria prophylaxis, brucellosis**	**Teeth discolouration & enamel hypoplasia, bone growth ↓ – avoid < 8 yrs, pregnancy, lactation**; **photosensitivity (sunscreen)**; **oesophageal ulcer – take with full glass of water, stay upright 30 min, not at bedtime**; **chelation – don't take with milk, antacids, iron, calcium** (2 hrs apart); **expired tetracycline – Fanconi syndrome**; superinfection; fatty liver (IV in pregnancy)
Chloramphenicol	Typhoid (earlier), meningitis (allergy), eye drops	**Gray baby syndrome (neonates)**, **bone marrow suppression – aplastic anaemia (idiosyncratic, fatal)**
Fluoroquinolones – ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, norfloxacin	**UTI, typhoid, gonorrhoea, Gram-negative**, **MDR-TB (levo/ moxi)**, traveller's diarrhoea, anthrax	**Tendinitis/ tendon rupture (Achilles – elderly, steroids)**, **arthropathy – avoid in children & pregnancy**, **QT prolongation**, CNS (seizures – with NSAIDs), photosensitivity, dysglycaemia; **chelation with antacids/ iron/ dairy**; ↑ theophylline
Sulfonamides – cotrimoxazole (sulfamethoxazole + trimethoprim)	**UTI, PCP (Pneumocystis) prophylaxis & Rx in HIV**, toxoplasmosis, nocardia	**SJS/ TEN (rash – stop)**, **crystalluria (drink plenty of fluids ≥ 2 L)**, **kernicterus (neonates, late pregnancy)**, haemolysis (G6PD), bone marrow suppression (folate ↓), hyperkalaemia; photosensitivity

Drug/ group	Key uses	Key adverse effects / nursing
Metronidazole (tinidazole, secnidazole)	**Anaerobes, amoebiasis (DOC), giardiasis, trichomoniasis, bacterial vaginosis, C. difficile, H. pylori, tetanus**, surgical prophylaxis	**Metallic taste**, nausea, **disulfiram-like reaction with alcohol (avoid alcohol during & 48–72 hrs after)**, dark urine, peripheral neuropathy (prolonged), **↑ warfarin effect**
Nitrofurantoin	**Uncomplicated UTI** (lower)	**Brown urine**, pulmonary fibrosis, neuropathy; avoid near term (haemolysis), G6PD
Vancomycin	**MRSA**, **C. difficile (oral)**, endocarditis	**"Red man (vancomycin flushing) syndrome" – infuse slowly over ≥ 60 min**; nephrotoxicity, ototoxicity; **trough levels**
Linezolid	MRSA, VRE, MDR-TB	Thrombocytopenia, myelosuppression, serotonin syndrome, optic neuropathy
Clindamycin	Anaerobes, bone, MRSA (skin)	**Pseudomembranous colitis (C. difficile) – report diarrhoea**
Polymyxin B / colistin	MDR Gram-negative (reserve)	Nephro- & neurotoxicity
Fosfomycin	UTI (single dose)	–

7.4 Antitubercular & Antileprosy Drugs

Drug	Key adverse effects
Isoniazid (H)	**Peripheral neuropathy (give pyridoxine 10–50 mg)**, **hepatitis**, slow acetylators; lupus-like; overdose seizures (pyridoxine antidote)
Rifampicin (R)	**Orange-red urine, sweat, tears (stains contact lenses)**, **hepatitis**, flu-like syndrome (intermittent), **potent enzyme inducer – OCP failure, ↓ warfarin, ↓ ART (protease inhibitors)**; take empty stomach
Pyrazinamide (Z)	**Hepatotoxicity (most hepatotoxic)**, **hyperuricaemia – gout/ arthralgia**
Ethambutol (E)	**Optic (retrobulbar) neuritis – ↓ visual acuity, red-green colour blindness** – check vision; caution in young children
Streptomycin (S)	**Ototoxicity (vestibular), nephrotoxicity**; avoid in pregnancy
Second-line (MDR-TB)	**Bedaquiline (QT)**, delamanid, pretomanid, linezolid, moxifloxacin/ levofloxacin, clofazimine, cycloserine (psychosis)

- **Regimen (NTEP) – 2HRZE + 4HRE (daily FDC, weight bands)**; **DOT/ 99DOTS**; **Ni-kshay Poshan Yojana ₹1000/month (2024)**; **TPT – INH 6 months/ 3HP**.
- **Leprosy MDT – rifampicin (monthly supervised) + clofazimine (skin pigmentation, ichthyosis) + dapsone (haemolysis – G6PD, methaemoglobinaemia, DRESS)**; **PB 6 months, MB 12 months**.

7.5 Antifungal Drugs

Drug	Points
Amphotericin B	**Systemic fungal infections (DOC – mucormycosis, cryptococcal meningitis), kala-azar**; **infusion reactions – fever, chills, rigors (premedicate – paracetamol, antihistamine, hydrocortisone)**; **nephrotoxicity (most important – hydrate with saline)**, **hypokalaemia, hypomagnesaemia**, anaemia; **liposomal – less toxic**; IV slowly in **5 % dextrose (not saline)**, **protect from light**
Azoles – fluconazole (candida, cryptococcal), **itraconazole, voriconazole** (aspergillosis – visual disturbances), **ketoconazole** (hepatotoxic, enzyme inhibitor), **posaconazole, isavuconazole**	**Enzyme inhibitors – drug interactions**; hepatotoxicity
Topical azoles – clotrimazole, miconazole	Skin, oral, vaginal candidiasis
Nystatin	**Oral thrush (swish & swallow), not absorbed – topical/ GI only**
Griseofulvin	**Dermatophytes (tinea capitis)**; **take with fatty meal**; long courses; enzyme inducer, photosensitivity, teratogenic
Terbinafine	**Onychomycosis (nail)**, tinea
Echinocandins – caspofungin, micafungin	Candidaemia

7.6 Antiviral & Antiretroviral Drugs

Drug	Use / points
Acyclovir, valacyclovir	**Herpes simplex, varicella-zoster, HSV encephalitis**; **nephrotoxicity (crystals) – infuse slowly, hydrate**
Ganciclovir, valganciclovir	**CMV**; bone marrow suppression
Oseltamivir	**Influenza (H1N1) – within 48 hrs**
Remdesivir, molnupiravir	COVID-19
Ribavirin	RSV, hepatitis C (old), haemolytic anaemia, **teratogenic**
Sofosbuvir + velpatasvir/ daclatasvir	**Hepatitis C (DAAs – cure)**
Tenofovir (TDF/ TAF), entecavir	**Hepatitis B**, HIV
Interferon	HBV/ HCV (old) – flu-like, depression

Antiretroviral therapy (ART)

- **Classes** – **NRTIs** (**zidovudine – anaemia, myopathy**; **tenofovir – nephrotoxicity, bone loss**; **lamivudine**, abacavir – hypersensitivity), **NNRTIs** (**nevirapine – rash/ SJS, hepatotoxicity; PPTCT infant prophylaxis**; **efavirenz – CNS (vivid dreams, dizziness), teratogenic (1st trimester)**),

protease inhibitors (lopinavir/ ritonavir, atazanavir – lipodystrophy, hyperglycaemia, jaundice), **integrase inhibitors** (**dolutegravir – preferred**; raltegravir).

- **First-line (NACO)** – **TLD (Tenofovir + Lamivudine + Dolutegravir)** – single pill daily; **"Test & Treat" – ART for all regardless of CD4 (2017)**.
- **Adherence > 95 %**; **viral load monitoring**; **IRIS (immune reconstitution inflammatory syndrome)**; **PEP – 28 days TLD, start within 72 hrs (ideally 2 hrs)**.
- **Cotrimoxazole prophylaxis**, **INH (TPT)**.

7.7 Antimalarial Drugs

Drug	Points
Chloroquine	**P. vivax (blood schizonticide) – 25 mg/kg over 3 days**; **pruritus (Africans), retinopathy (long-term – eye checks)**, GI upset, QT; also amoebic liver abscess, RA/ SLE
Primaquine	**Tissue schizonticide – radical cure of vivax (hypnozoites) – 0.25 mg/kg × 14 days**; **gametocidal – falciparum single dose 0.75 mg/kg**; **haemolysis in G6PD deficiency**; **contraindicated in pregnancy & infants < 1 yr**; give with food
Artemisinins – artesunate, artemether	**Falciparum (fastest acting)**; **ACT – artesunate + sulfadoxine-pyrimethamine (India, except NE) / artemether-lumefantrine (NE states)**; **injectable artesunate – severe malaria DOC**
Quinine	Severe malaria (alternative), 1st trimester; **cinchonism (tinnitus, deafness, headache, vertigo)**, **hypoglycaemia (↑ insulin)**, QT; IV slowly in dextrose
Mefloquine	Prophylaxis; neuropsychiatric effects
Doxycycline	Prophylaxis (travellers), with quinine
Atovaquone-proguanil	Prophylaxis

7.8 Anthelmintics & Antiprotozoals

Drug	Use
Albendazole	**Broad – roundworm, hookworm, whipworm, pinworm, tapeworm, neurocysticercosis, hydatid**; **400 mg single dose (200 mg 1–2 yrs)**; National Deworming Day; avoid 1st trimester
Mebendazole	100 mg BD × 3 days; pinworm (single dose, repeat 2 weeks)
Pyrantel pamoate	Roundworm, pinworm (safe in pregnancy)
Ivermectin	**Strongyloides, scabies (oral), filaria (MDA – IDA), onchocerciasis**
Praziquantel	**Tapeworms, schistosomiasis**, neurocysticercosis
DEC (diethylcarbamazine)	**Filariasis (MDA with albendazole)**, tropical eosinophilia
Niclosamide	Tapeworm
Metronidazole/ tinidazole	Amoebiasis, giardia, trichomonas
Diloxanide furoate	Luminal amoebicide (cyst passers)

Drug	Use
Sodium stibogluconate, miltefosine, liposomal amphotericin B	**Kala-azar** (single-dose L-AmB – India)
Permethrin 5 % (topical)	**Scabies (DOC)**; 1 % – head lice
Benzyl benzoate 25 %	Scabies

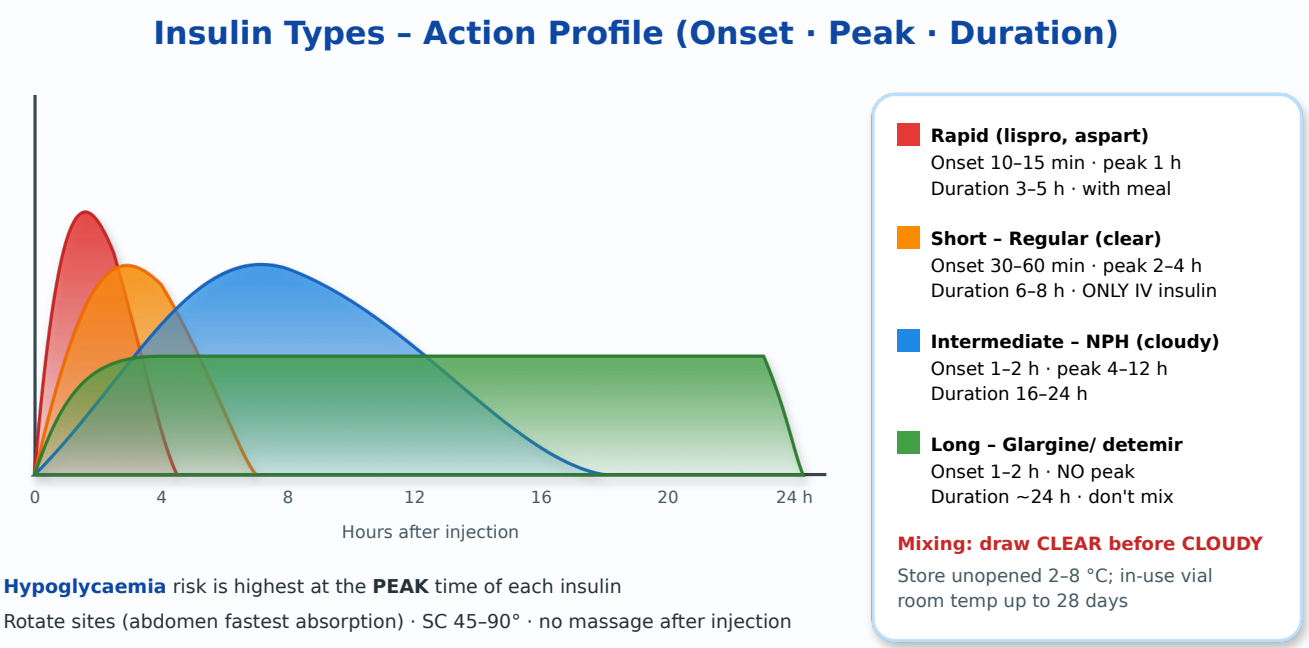
PYQ style

- Penicillin discovered by – **Alexander Fleming (1928)**; most dangerous reaction – **anaphylaxis**.
- Before penicillin – **ask allergy history, test dose, keep adrenaline ready**.
- Rheumatic fever prophylaxis – **benzathine penicillin every 21 days**.
- Penicillinase-resistant – **cloxacillin**; β-lactamase inhibitor – **clavulanic acid**.
- Surgical prophylaxis – **cefazolin**; meningitis/ typhoid – **ceftriaxone**.
- Aminoglycoside toxicity – **nephrotoxicity & ototoxicity**.
- Tetracyclines – **teeth staining (avoid < 8 yrs, pregnancy)**; take with – **full glass of water, upright, not with milk**.
- Scrub typhus DOC – **doxycycline**.
- Gray baby syndrome – **chloramphenicol**; aplastic anaemia – **chloramphenicol**.
- Fluoroquinolone – **tendon rupture**; avoid in **children**.
- Cotrimoxazole – **crystalluria (fluids), SJS**; PCP prophylaxis in HIV.
- Metronidazole + alcohol – **disulfiram-like reaction**; taste – **metallic**.
- Amoebiasis DOC – **metronidazole**.
- Red man syndrome – **vancomycin (infuse slowly over 60 min)**; MRSA – **vancomycin**.
- Clindamycin – **pseudomembranous colitis**.
- INH – **peripheral neuropathy (pyridoxine)**; rifampicin – **orange urine**; ethambutol – **optic neuritis**; pyrazinamide – **gout**.
- Amphotericin B – **nephrotoxicity, infusion fever/ chills**; dilute in **5 % dextrose**.
- Oral thrush – **nystatin**; griseofulvin – **with fatty meal**.
- Herpes – **acyclovir**; influenza – **oseltamivir**; CMV – **ganciclovir**.
- HIV first-line – **TLD**; zidovudine – **anaemia**; tenofovir – **nephrotoxicity**; nevirapine – **rash, hepatotoxicity**.
- Vivax radical cure – **primaquine 14 days**; contraindicated in – **G6PD, pregnancy**.
- Severe malaria – **injectable artesunate**; quinine – **cinchonism, hypoglycaemia**.
- Chloroquine long-term – **retinopathy**.
- Broad anthelmintic – **albendazole**; filaria – **DEC**; tapeworm/ schistosomiasis – **praziquantel**.
- Scabies – **permethrin 5 %**.

8 Endocrine Drugs & Hormones

8.1 Insulin

- **Discovered – Banting & Best (1921)**; **human/ analogue insulin (recombinant)**; **U-100 (100 units/ml)**.
- **Indications** – **type 1 DM (essential)**, type 2 DM (uncontrolled), **gestational DM (DOC)**, **DKA, HHS** (IV regular), surgery/ critical illness, **hyperkalaemia (with glucose)**.



Insulin onset, peak & duration – rapid, short, intermediate & long-acting

Type	Examples	Onset	Peak	Duration	Points
Rapid-acting	**Lispro, aspart, glulisine**	**10–15 min**	**~1 hr**	3–5 hrs	**Give with meal (0–15 min before)**
Short-acting (regular/ soluble)	**Regular (Actrapid) – clear**	**30–60 min**	**2–4 hrs**	6–8 hrs	**30 min before meals**; **only insulin given IV (DKA)**
Intermediate	**NPH/ isophane (Insulatard) – cloudy**	1–2 hrs	**4–12 hrs**	16–24 hrs	Roll gently (don't shake)
Long-acting (basal)	**Glargine, detemir, degludec**	1–2 hrs	**No peak**	**~24 hrs (degludec > 40)**	Once daily same time; **don't mix with other insulins**
Premixed	30/70 (30 % regular + 70 % NPH), 50/50	–	Dual	–	Twice daily before meals

- **Mixing – draw CLEAR (regular) before CLOUDY (NPH)** ("**RN – Regular then NPH**"); inject air into NPH first, then regular.
- **Sites – abdomen (fastest absorption – except 5 cm around umbilicus)**, thigh, upper arm, buttocks; **rotate within a region** (prevent **lipohypertrophy/ lipodystrophy**); **SC at 90° (45° if thin)**; **don't**

massage; insulin pen – hold 10 sec after injection.

- **Storage** – **unopened 2–8 °C (refrigerator, not freezer)**; **in-use vial/ pen at room temperature (< 30 °C) for ~28 days**; protect from heat/ sunlight; **don't use if frozen/ clumped**.
- **Side effects** – **hypoglycaemia (most common & dangerous)**, **weight gain**, lipodystrophy, **hypokalaemia (IV insulin)**, allergy, insulin resistance; **Somogyi effect** (night hypoglycaemia → rebound morning hyperglycaemia – ↓ evening insulin) vs **dawn phenomenon** (early morning ↑ GH/ cortisol – ↑ evening dose/ shift timing).
- **DKA** – **IV regular insulin 0.1 U/kg/hr** (after fluids, $K^+ \geq 3.3$); add dextrose when glucose ~ 250 mg/dl; **monitor K^+** (insulin drives K^+ into cells).
- **Sliding scale** – regular insulin by blood glucose readings.

Hypoglycaemia

- **Blood glucose < 70 mg/dl**; S/S – **adrenergic (sweating – most common, tremor, palpitations, hunger, anxiety)**, **neuroglycopenic (confusion, irritability, blurred vision, slurred speech, seizures, coma)**; **β-blockers mask adrenergic signs (except sweating)**.
- **Conscious – "Rule of 15": 15–20 g fast-acting carbohydrate** (glucose tablets, ½ cup juice, 3–4 tsp sugar, regular soft drink), **recheck after 15 min**, repeat if < 70, then complex carbohydrate snack.
- **Unconscious – 25–50 ml of 50 % dextrose IV (25 % in children; 10 % 2–5 ml/kg in neonates/ children)** or **glucagon 1 mg IM/ SC** (**0.5 mg < 25 kg**); **nothing by mouth**; **sulfonylurea-induced – prolonged (observe 24–72 hrs, octreotide)**.

8.2 Oral & Non-Insulin Antidiabetic Drugs

Class	Drugs	Mechanism	Key points
Biguanide	**Metformin**	**↓ hepatic glucose production**, ↑ insulin sensitivity	**First-line T2DM**; **no hypoglycaemia (alone), weight neutral/ loss**; **GI upset (give with meals)**, metallic taste, **B12 deficiency**, **lactic acidosis (rare – stop in renal failure eGFR < 30, contrast, surgery, hypoxia, alcohol)**; PCOS
Sulfonylureas	**Glibenclamide (glyburide), glimepiride, gliclazide, glipizide**	↑ insulin release	**Hypoglycaemia (esp. glibenclamide, elderly, renal)**, **weight gain**; give before breakfast; disulfiram-like (chlorpropamide)
Meglitinides	Repaglinide, nateglinide	↑ insulin (short)	Take with each meal; skip if meal skipped
DPP-4 inhibitors ("-gliptin")	**Sitagliptin, vildagliptin, teneligliptin**	↑ incretin	Weight neutral, low hypo risk; pancreatitis rare
SGLT2 inhibitors ("-gliflozin")	**Dapagliflozin, empagliflozin, canagliflozin**	**↑ urinary glucose excretion**	**Genital mycotic infections, UTI**, volume depletion, **euglycaemic DKA**; **cardio-renal benefits (HF, CKD)**
Thiazolidinediones	**Pioglitazone**	↑ insulin sensitivity (PPAR-γ)	**Fluid retention, weight gain – avoid in heart failure**; bladder cancer, fractures

Class	Drugs	Mechanism	Key points
α-glucosidase inhibitors	**Acarbose**, voglibose, miglitol	↓ carbohydrate absorption	**Flatulence, diarrhoea**; take with first bite; treat hypo with **glucose (not sucrose)**
GLP-1 receptor agonists (injectable/ oral)	**Liraglutide, semaglutide, dulaglutide, exenatide**	↑ insulin, ↓ glucagon, slow gastric emptying, satiety	**Weight loss**, nausea, pancreatitis; CV benefit

8.3 Thyroid & Antithyroid Drugs

Drug	Points
Levothyroxine (T4)	**Hypothyroidism/ myxoedema, cretinism (10–15 µg/kg)**; **take on empty stomach, 30–60 min before breakfast, same time daily**; **separate from iron, calcium, antacids (4 hrs)**; **lifelong**; overdose → hyperthyroid symptoms (palpitations, weight loss, AF, osteoporosis); **start low in elderly/ cardiac patients (angina)**; monitor TSH (6–8 weeks)
Liothyronine (T3)	Myxoedema coma (IV)
Carbimazole / methimazole	**Hyperthyroidism (DOC)**; **agranulocytosis – report sore throat & fever (stop drug, check WBC)**, rash; **teratogenic (aplasia cutis) – avoid 1st trimester**
Propylthiouracil (PTU)	**Thyroid storm & 1st trimester pregnancy**; ↓ T4 → T3 conversion; **hepatotoxicity**
Radioactive iodine (I-131)	Definitive Rx (Graves', toxic nodule); **contraindicated in pregnancy & lactation**; **hypothyroidism (common late)**; isolation precautions, avoid close contact for days
Lugol's iodine / potassium iodide	**Pre-operative (↓ vascularity of gland, 10 days)**, thyroid storm (after thionamide), **radiation emergency (protects thyroid)**
Propranolol	Symptom control (tremor, tachycardia)

8.4 Corticosteroids

- **Glucocorticoids** – **hydrocortisone** (short, + mineralocorticoid – adrenal crisis), **prednisolone** (intermediate, oral), **methylprednisolone** (IV pulse), **dexamethasone & betamethasone** (long-acting, potent, no mineralocorticoid – **cerebral oedema, fetal lung maturity**, antiemetic); **fludrocortisone** (mineralocorticoid – Addison's).
- **Uses** – inflammation/ allergy (**anaphylaxis adjunct, asthma**, urticaria), autoimmune (SLE, RA, IBD), **nephrotic syndrome**, organ transplant, cancer (lymphoma, ALL), **adrenal insufficiency (replacement)**, **cerebral oedema (dexamethasone)**, **antenatal (lung maturity)**, shock (septic – hydrocortisone), skin (topical), eye.
- **Adverse effects (long-term) – "CUSHINGOID"** – **Cushingoid features (moon face, buffalo hump, truncal obesity, striae)**, **hyperglycaemia (steroid diabetes)**, **hypertension, fluid retention**, **hypokalaemia**, **osteoporosis & fractures, avascular necrosis**, **immunosuppression – infections (TB reactivation, fungal), delayed wound healing**, **peptic ulcer (with NSAIDs)**, **cataract, glaucoma**, **growth suppression (children)**, myopathy, **psychosis/ mood changes/ insomnia**, skin thinning, acne, hirsutism, **HPA axis suppression → adrenal crisis on abrupt withdrawal**.
- **Nursing/ teaching** – **give in the morning (mimic diurnal rhythm) with food/ milk** (GI), **never stop abruptly – taper**, **steroid card**, **stress dose ↑** (surgery, illness), monitor **BP, weight, glucose, K^+**,

infection signs, eye check, calcium + vitamin D, low-salt high-protein diet, **avoid live vaccines** (high dose); **ICS – rinse mouth**.

- **Adrenal crisis** – hypotension, shock, hypoglycaemia → **IV hydrocortisone 100 mg + NS/ dextrose**.

8.5 Sex Hormones, Contraceptives & Uterine Drugs

- **Oestrogens** (estradiol, ethinylestradiol) – HRT, contraception; **DVT/ PE, breast & endometrial cancer (unopposed), stroke**, nausea, breast tenderness.
- **Progestins** (medroxyprogesterone, levonorgestrel, norethisterone) – contraception, AUB, endometriosis.
- **Combined OCP** (Mala-N) – **inhibits ovulation**; **contraindications – smoker > 35 yrs, DVT history, breast cancer, liver disease, migraine with aura, uncontrolled HTN, < 6 months postpartum breastfeeding**; **warning signs "ACHES"** – Abdominal pain, Chest pain, Headache (severe), Eye (visual) problems, Severe leg pain; **missed pill** – take ASAP; **enzyme inducers (rifampicin, phenytoin, carbamazepine) & some antibiotics ↓ efficacy**.
- **Emergency contraception** – **levonorgestrel 1.5 mg within 72 hrs**; **ulipristal**; Cu-IUD within 5 days.
- **Antiprogestin** – **mifepristone (RU-486) – medical abortion with misoprostol**.
- **Centchroman (ormeloxifene – Chhaya)** – SERM, weekly.
- **Clomiphene citrate, letrozole** – ovulation induction (PCOS).
- **Tamoxifen** – **SERM – breast cancer (ER+)**; **endometrial cancer, DVT, hot flushes**. **Raloxifene** – osteoporosis.
- **Testosterone** – hypogonadism; **finasteride (5α-reductase inhibitor) – BPH**, male-pattern baldness; teratogenic – pregnant women shouldn't handle crushed tablets.
- **Uterine stimulants (oxytocics)** – **oxytocin** (induction, PPH – water intoxication), **methylergometrine** (PPH – contraindicated HTN), **carboprost** (PGF2α – asthma contraindicated), **misoprostol** (PGE1), **dinoprostone** (PGE2 – cervical ripening).
- **Tocolytics (uterine relaxants)** – **nifedipine (first-line)**, **terbutaline/ ritodrine (β2)**, atosiban, indomethacin, **$MgSO_4$** (neuroprotection).
- **Antenatal steroids** – dexamethasone 6 mg IM × 4 (12-hrly) / betamethasone 12 mg × 2.

8.6 Pituitary, Bone & Other Hormones

Drug	Use / point
Vasopressin (ADH)/ desmopressin (DDAVP)	**Diabetes insipidus (DOC – desmopressin intranasal/ oral)**, **nocturnal enuresis**, **haemophilia A (mild)/ von Willebrand**, variceal bleeding (vasopressin/ terlipressin); **hyponatraemia, water intoxication**
Octreotide (somatostatin analogue)	**Acromegaly, carcinoid, variceal bleeding**, sulfonylurea overdose
Growth hormone (somatropin)	GH deficiency
Bromocriptine, cabergoline	**Hyperprolactinaemia**, suppress lactation, acromegaly, Parkinson's

Drug	Use / point
Bisphosphonates – alendronate (weekly), zoledronic acid (yearly IV)	**Osteoporosis, Paget's, hypercalcaemia of malignancy**; **alendronate – empty stomach, with full glass of plain water, stay upright 30 min (oesophagitis)**; osteonecrosis of jaw
Calcium + vitamin D (cholecalciferol)	Osteoporosis, rickets; calcitriol in CKD
Calcitonin	Hypercalcaemia, Paget's
Gout – allopurinol (xanthine oxidase inhibitor – **prophylaxis; not in acute attack; SJS (HLA-B*5801); fluids**), **febuxostat**, **colchicine (acute – diarrhoea)**, NSAIDs (acute), **probenecid** (uricosuric), **rasburicase** (tumour lysis)	–

PYQ style

- Insulin discovered – **Banting & Best (1921)**; U-100 – **100 units/ml**.
- Only insulin given IV – **regular (short-acting)**.
- Peakless insulin – **glargine**; cloudy insulin – **NPH**.
- Mixing – **clear (regular) before cloudy (NPH)**.
- Fastest absorption site – **abdomen**; rotate sites – **prevent lipodystrophy**.
- Unopened insulin storage – **2–8 °C**; in-use – **room temp up to 28 days**.
- Most common insulin side effect – **hypoglycaemia**.
- Hypoglycaemia conscious – **15 g sugar, recheck in 15 min**; unconscious – **IV 50 % dextrose / glucagon IM**.
- β-blockers mask hypoglycaemia except – **sweating**.
- Somogyi effect – **night hypoglycaemia → morning hyperglycaemia**.
- First-line T2DM – **metformin**; serious side effect – **lactic acidosis**; stop before – **contrast**.
- Hypoglycaemia & weight gain – **sulfonylureas (glibenclamide)**.
- SGLT2 inhibitors – **genital infections, UTI**.
- Pioglitazone avoided in – **heart failure**.
- Acarbose side effect – **flatulence**.
- Levothyroxine – **empty stomach, morning, lifelong**.
- Carbimazole serious side effect – **agranulocytosis (sore throat)**.
- Antithyroid in 1st trimester & thyroid storm – **PTU**.
- Radioactive iodine – **contraindicated in pregnancy**.
- Lugol's iodine pre-op – **↓ vascularity**.
- Steroids given – **morning with food; taper, never stop abruptly**.
- Steroid long-term side effects – **hyperglycaemia, osteoporosis, infection, Cushingoid, peptic ulcer**.
- Adrenal crisis – **IV hydrocortisone**.
- OCP warning signs – **ACHES**.
- Emergency contraception – **levonorgestrel 1.5 mg within 72 hrs**.
- Tamoxifen side effect – **endometrial cancer, DVT**.
- Diabetes insipidus DOC – **desmopressin**.
- Alendronate – **empty stomach, full glass water, upright 30 min**.
- Allopurinol – **not in acute gout**; acute gout – **colchicine/ NSAIDs**.

9 Anticancer Drugs, Immunopharmacology, Antidotes & Special Populations

9.1 Anticancer (Antineoplastic) Drugs

Class	Drugs	Key toxicities
Alkylating agents	**Cyclophosphamide, ifosfamide**, chlorambucil, busulfan, melphalan	**Cyclophosphamide/ ifosfamide – haemorrhagic cystitis (prevent: hydration, frequent voiding, MESNA)**; busulfan – pulmonary fibrosis; alopecia, myelosuppression, infertility
Platinum compounds	**Cisplatin**, carboplatin, oxaliplatin	**Cisplatin – nephrotoxicity (hydration, Mg), ototoxicity, peripheral neuropathy, most emetogenic (severe vomiting)**; oxaliplatin – cold-induced neuropathy
Antimetabolites	**Methotrexate** (folate antagonist), **5-fluorouracil**, 6-mercaptopurine, cytarabine, gemcitabine, capecitabine	**Methotrexate – mucositis, myelosuppression, hepatotoxicity, pulmonary fibrosis; antidote – leucovorin (folinic acid) rescue**; **weekly dose for RA/ psoriasis (fatal error if daily)**; 6-MP – interacts with allopurinol (↓ dose)
Plant alkaloids – vinca	**Vincristine** (**peripheral neuropathy, constipation, paralytic ileus – NOT myelosuppressive; fatal if given intrathecally – IV only**), **vinblastine (myelosuppression)**	**Vesicants – extravasation**
Taxanes	Paclitaxel, docetaxel	Hypersensitivity (premedicate), neuropathy
Topoisomerase inhibitors	Etoposide, irinotecan (diarrhoea)	–
Antitumour antibiotics	**Doxorubicin (Adriamycin), daunorubicin – cardiotoxicity (cumulative dose – cardiomyopathy, CHF; dexrazoxane), red urine**; **bleomycin – pulmonary fibrosis**, skin pigmentation; **actinomycin D**	Vesicant
Hormonal	**Tamoxifen** (breast), aromatase inhibitors (**letrozole, anastrozole**), **leuprolide/ goserelin** (prostate), **flutamide/ bicalutamide**, prednisolone	Hot flushes, osteoporosis
Others	**L-asparaginase** (ALL – **anaphylaxis, pancreatitis**), hydroxyurea (sickle cell, CML), **imatinib** (CML – targeted), **rituximab, trastuzumab (cardiotoxic), bevacizumab** (monoclonals)	–

- **Common toxicities** – **bone marrow suppression (most common dose-limiting – neutropenia nadir 7–14 days)**, **nausea/ vomiting (ondansetron before)**, **alopecia (reversible)**, **mucositis/ stomatitis**, diarrhoea, **infertility**, teratogenicity, **tumour lysis syndrome** (hyperuricaemia, hyperkalaemia, hyperphosphataemia, hypocalcaemia → AKI – **hydration, allopurinol/ rasburicase**), extravasation, secondary malignancy.
- **Nursing** – **neutropenic precautions** (hand hygiene, **no fresh flowers/ raw food**, report **fever ≥ 38 °C**), bleeding precautions (platelets), **mouth care (soft brush, saline rinses)**, nutrition, hydration, antiemetics before chemo, **check CBC before each cycle**, **extravasation – stop infusion, aspirate, don't flush, elevate, cold compress (most)/ warm (vinca alkaloids), antidote (dexrazoxane – anthracyclines; hyaluronidase – vinca)**.
- **Safe handling (cytotoxic precautions)** – **biological safety cabinet (class II)**, **chemo-resistant gloves (double), impermeable gown, mask/ goggles**, **luer-lock syringes, closed systems**, **cytotoxic waste – yellow bag with cytotoxic symbol (BMW)**, spill kit, **pregnant/ breastfeeding staff should avoid preparation**, patient's body fluids (**48 hrs** after chemo – gloves; double flush toilet).

9.2 Immunopharmacology

Group	Drugs	Points
Calcineurin inhibitors	**Cyclosporine, tacrolimus**	**Transplant rejection**; **nephrotoxicity**, HTN, **gum hyperplasia & hirsutism (cyclosporine)**, hyperglycaemia/ neurotoxicity (tacrolimus); **TDM**; grapefruit ↑ levels
Antimetabolites	**Azathioprine**, mycophenolate	Myelosuppression
mTOR inhibitors	Sirolimus, everolimus	–
Corticosteroids	Prednisolone	–
DMARDs (RA)	**Methotrexate (anchor – weekly + folic acid)**, hydroxychloroquine (retinopathy – eye check), sulfasalazine, leflunomide; **biologics – infliximab, adalimumab, etanercept (anti-TNF – TB reactivation – screen before)**	–

- **Vaccines & sera** – **active immunity (vaccines – live/ killed/ toxoid)**; **passive (antisera/ immunoglobulins)** – **ATS (equine anti-tetanus serum – test dose)/ TIG (human – preferred)**, **ARS/ RIG**, **ADS (anti-diphtheritic serum)**, **ASV (polyvalent anti-snake venom – cobra, krait, Russell's viper, saw-scaled viper; test dose not recommended now; watch anaphylaxis – adrenaline ready)**, **HBIG**, **anti-D**, **IVIG**.
- **Serum sickness** – type III (7–14 days after antiserum) – fever, rash, arthralgia.

9.3 Vitamins (therapeutic use)

Vitamin	Use / toxicity
A	Xerophthalmia (prophylaxis 1–2 lakh IU); **toxicity – raised ICP (pseudotumour), teratogenic (isotretinoin)**
D	Rickets, osteoporosis, hypocalcaemia; **toxicity – hypercalcaemia**

Vitamin	Use / toxicity
K (phytonadione)	**Warfarin reversal, newborn (1 mg IM)**; IV slowly (anaphylactoid)
B1 (thiamine)	**Wernicke's (before glucose), beriberi, alcoholics**
B6 (pyridoxine)	**INH neuropathy prevention**, INH overdose seizures, morning sickness
B12	Pernicious anaemia (IM), **cyanide poisoning (hydroxocobalamin)**
Folic acid	Megaloblastic anaemia, NTD prevention, **methotrexate toxicity (folinic acid)**
Niacin	Pellagra, dyslipidaemia (flushing)
C	Scurvy, ↑ iron absorption, methaemoglobinaemia

9.4 Antidotes – Master Table

Poison / drug	Antidote
Paracetamol	**N-acetylcysteine (NAC)**
Opioids (morphine, heroin)	**Naloxone**
Benzodiazepines	**Flumazenil**
Heparin	**Protamine sulphate**
Warfarin	**Vitamin K** (+ FFP/ PCC)
Dabigatran	**Idarucizumab**
Thrombolytics	**Tranexamic acid / aminocaproic acid**
Organophosphates	**Atropine + pralidoxime (2-PAM)**
Atropine/ anticholinergics (datura)	**Physostigmine**
Digoxin	**Digoxin-specific Fab (Digibind)**
Iron	**Desferrioxamine**
Lead	**$CaNa_2EDTA$, DMSA (succimer), BAL**
Arsenic, mercury, gold	**BAL (dimercaprol)**, DMSA
Copper (Wilson's)	**Penicillamine**
Methotrexate	**Leucovorin (folinic acid)**
Cyclophosphamide (bladder)	**MESNA**
Doxorubicin extravasation/ cardiotoxicity	**Dexrazoxane**
Isoniazid	**Pyridoxine**
Magnesium sulphate	**Calcium gluconate**
Insulin/ sulfonylurea hypoglycaemia	**Dextrose, glucagon (octreotide – SU)**
β-blockers	**Glucagon**
Calcium channel blockers	**Calcium gluconate/ chloride**, high-dose insulin
Cyanide (nitroprusside)	**Hydroxocobalamin**; sodium nitrite + sodium thiosulphate
Methanol / ethylene glycol	**Fomepizole / ethanol**

Poison / drug	Antidote
Carbon monoxide	**100 % oxygen / hyperbaric O_2**
Methaemoglobinaemia (nitrites, dapsone, prilocaine)	**Methylene blue**
Local anaesthetic toxicity	**20 % lipid emulsion**
Tricyclic antidepressants	**Sodium bicarbonate**
Malignant hyperthermia / NMS	**Dantrolene**
Serotonin syndrome	**Cyproheptadine**
Snake bite	**Polyvalent ASV**; neostigmine + atropine (neurotoxic)
Scorpion sting	**Prazosin**
Acute dystonia (metoclopramide/ antipsychotics)	**Promethazine/ diphenhydramine/ benztropine**
Theophylline	Activated charcoal (multi-dose), haemodialysis
Lithium	**Haemodialysis**
Salicylates (aspirin)	**Sodium bicarbonate (urine alkalinisation)**, haemodialysis
Hyperkalaemia	**Calcium gluconate** (stabilise), insulin-glucose
Kerosene	No emesis; supportive
General oral poisoning	**Activated charcoal 1 g/kg within 1 hr** (not for iron, lithium, alcohols, kerosene, corrosives)

9.5 Drugs in Special Populations

Pregnancy

- **FDA categories (old)** – **A** (safe in humans – folic acid, thyroxine), **B** (animal safe – penicillins, cephalosporins, paracetamol, metformin, insulin), **C** (risk not ruled out), **D** (evidence of risk, benefit may justify – phenytoin, tetracycline, lithium), **X (contraindicated – risk outweighs benefit)** – **isotretinoin, thalidomide, warfarin (first trimester), statins, methotrexate, misoprostol, ribavirin, finasteride, leflunomide, diethylstilbestrol**; **(FDA replaced categories with PLLR narrative labelling in 2015)**.
- **Critical period – organogenesis (3–8 weeks)**.

Teratogen	Defect
Thalidomide	**Phocomelia**
Warfarin	Nasal hypoplasia, stippled epiphyses
Valproate	**Neural tube defects** (spina bifida)
Phenytoin	Fetal hydantoin syndrome (cleft lip, digit hypoplasia)
Isotretinoin (vitamin A)	Craniofacial, cardiac, CNS defects
Lithium	**Ebstein's anomaly**
ACE inhibitors/ ARBs	Fetal renal failure, oligohydramnios

Teratogen	Defect
Tetracyclines	Teeth discolouration, bone
Aminoglycosides (streptomycin)	**Ototoxicity (deafness)**
DES (diethylstilbestrol)	Vaginal adenocarcinoma in daughters
Methotrexate	Abortion, skull defects
Alcohol	**Fetal alcohol syndrome**
Carbimazole	Aplasia cutis
Misoprostol	Abortion, Möbius
Androgens/ danazol	Virilisation of female fetus

- **Safe** – **penicillins, cephalosporins, erythromycin/ azithromycin, paracetamol, insulin, heparin/ LMWH, methyldopa, labetalol, nifedipine, levothyroxine, folic acid, iron**.
- **Lactation** – most drugs safe; **avoid – cytotoxics, radioactive iodine, lithium, amiodarone, ergot alkaloids/ bromocriptine (suppress lactation), tetracyclines (prolonged), chloramphenicol, high-dose oestrogen OCP**.

Children & elderly

- **Children** – dose by **weight/ BSA**; immature liver & kidney (**gray baby – chloramphenicol, kernicterus – sulfonamides**); **avoid aspirin (Reye's), tetracyclines (< 8 yrs), fluoroquinolones, codeine (< 12 yrs), loperamide (< 2–4 yrs)**; liquid formulations; calibrated syringe.
- **Elderly** – ↓ renal/ hepatic function, ↓ albumin, ↑ body fat → **↑ sensitivity & toxicity**; **"start low, go slow"**; **polypharmacy** – review; **Beers criteria** (potentially inappropriate drugs – **long-acting BZD, anticholinergics, first-gen antihistamines, NSAIDs long-term, digoxin > 0.125 mg**); falls (sedatives, antihypertensives); confusion; compliance aids (pill organisers).

Renal & hepatic disease

- **Renal impairment** – adjust dose of **aminoglycosides, vancomycin, digoxin, lithium, metformin (stop if eGFR < 30), LMWH, gabapentin**; **avoid NSAIDs, nephrotoxins**.
- **Liver disease** – avoid/ reduce **hepatotoxic & highly metabolised drugs (paracetamol high dose, opioids, BZD – prefer lorazepam/ oxazepam)**; **↓ albumin → ↑ free drug**; bleeding risk.

9.6 Important Drug Interactions

Combination	Result
Warfarin + aspirin/ NSAIDs / metronidazole / cotrimoxazole / fluconazole / amiodarone	**↑ bleeding**
Rifampicin + OCP / warfarin / ART	**↓ effect (enzyme induction)**
MAOI + tyramine foods / SSRIs / pethidine	Hypertensive crisis / serotonin syndrome
Digoxin + diuretics (hypokalaemia)	**Digoxin toxicity**
ACEI + K-sparing diuretics / K supplements	**Hyperkalaemia**
Lithium + thiazides / NSAIDs / ACEI	**Lithium toxicity**

Combination	Result
Nitrates + sildenafil	**Severe hypotension**
Metronidazole / cephalosporins + alcohol	Disulfiram-like reaction
Tetracycline / ciprofloxacin + antacids / iron / milk	**↓ absorption (chelation)**
Aminoglycoside + frusemide	↑ ototoxicity
Theophylline + ciprofloxacin / erythromycin	Theophylline toxicity
Allopurinol + azathioprine / 6-MP	Bone marrow toxicity
Levodopa + pyridoxine	↓ levodopa effect
Sildenafil + nitrates	Hypotension
Grapefruit + statins / CCBs / cyclosporine	↑ levels
Alcohol + BZD/ opioids	Respiratory depression

PYQ style

- Haemorrhagic cystitis – **cyclophosphamide**; prevention – **MESNA + hydration**.
- Cardiotoxic anticancer drug – **doxorubicin**; red urine – **doxorubicin**.
- Pulmonary fibrosis – **bleomycin (also busulfan, methotrexate, amiodarone)**.
- Vincristine – **peripheral neuropathy, never intrathecal**.
- Most emetogenic – **cisplatin**; nephrotoxic & ototoxic – **cisplatin**.
- Methotrexate antidote – **leucovorin**.
- Commonest dose-limiting chemo toxicity – **bone marrow suppression**.
- Chemo safe handling – **biological safety cabinet, double gloves, gown**.
- Cyclosporine – **nephrotoxicity, gum hyperplasia**.
- Anti-TNF drugs – **screen for TB**.
- Heparin antidote – **protamine**; warfarin – **vitamin K**.
- Iron poisoning – **desferrioxamine**; lead – **EDTA**; arsenic/ mercury – **BAL**; copper – **penicillamine**.
- Cyanide – **hydroxocobalamin**; methanol – **fomepizole/ ethanol**.
- Methaemoglobinaemia – **methylene blue**.
- Activated charcoal dose – **1 g/kg**; not for – **iron, lithium, kerosene**.
- Pregnancy category X – **isotretinoin, thalidomide, statins, warfarin**.
- Valproate – **NTD**; lithium – **Ebstein's**; streptomycin – **deafness**.
- Safe antihypertensive in pregnancy – **methyldopa/ labetalol**.
- Elderly prescribing – **start low, go slow; Beers criteria**.
- Metformin stopped in – **eGFR < 30, contrast**.
- Rifampicin + OCP – **contraceptive failure**.
- Nitrates + sildenafil – **severe hypotension**.
- Tetracycline + milk – **chelation (↓ absorption)**.

10 Quick Revision – Master Tables & 150 One-Liners

10.1 Drug Name Suffixes (Stem Recognition)

Suffix	Group	Example
-olol	β-blocker	Propranolol, atenolol, metoprolol
-pril	ACE inhibitor	Enalapril, captopril, ramipril
-sartan	ARB	Losartan, telmisartan
-dipine	Dihydropyridine CCB	Amlodipine, nifedipine
-statin	HMG-CoA reductase inhibitor	Atorvastatin, rosuvastatin
-prazole	Proton pump inhibitor	Omeprazole, pantoprazole
-tidine	H2 blocker	Famotidine
-azosin	α1-blocker	Prazosin, terazosin (tamsulosin also α1)
-terol	β2 agonist	Salmeterol, formoterol
-pam / -lam	Benzodiazepine	Diazepam, lorazepam, midazolam
-cillin	Penicillin	Amoxicillin, cloxacillin
cef- / ceph-	Cephalosporin	Ceftriaxone, cephalexin
-mycin / -micin	Aminoglycoside / macrolide	Gentamicin, erythromycin
-floxacin	Fluoroquinolone	Ciprofloxacin
-cycline	Tetracycline	Doxycycline
-azole	Antifungal (also metronidazole)	Fluconazole
-vir	Antiviral	Acyclovir, tenofovir
-gliptin / -gliflozin / -glitazone	DPP-4 / SGLT2 / TZD	Sitagliptin, dapagliflozin, pioglitazone
-parin	Heparin (LMWH)	Enoxaparin
-xaban / -gatran	Factor Xa / thrombin inhibitor	Rivaroxaban, dabigatran
-plase	Thrombolytic	Alteplase, tenecteplase
-tropium	Anticholinergic (inhaled)	Ipratropium, tiotropium
-setron	5-HT3 antagonist	Ondansetron
-triptan	5-HT1 agonist (migraine)	Sumatriptan
-curonium	Non-depolarising NM blocker	Vecuronium, rocuronium
-caine	Local anaesthetic	Lignocaine, bupivacaine
-sone / -olone	Corticosteroid	Hydrocortisone, prednisolone
-mab	Monoclonal antibody	Rituximab, trastuzumab
-nib	Tyrosine kinase inhibitor	Imatinib

Suffix	Group	Example
-dronate	Bisphosphonate	Alendronate
-oxetine / -aline / -opram	SSRI	Fluoxetine, sertraline, escitalopram

10.2 Drugs of Choice (DOC)

Condition	DOC
Anaphylaxis	**Adrenaline 0.5 mg IM (1:1000)**
Cardiac arrest (VF/ asystole)	**Adrenaline 1 mg IV (1:10,000)**
Acute asthma	**Salbutamol (nebulised)**
Symptomatic bradycardia	**Atropine**
PSVT	**Adenosine** (after vagal manoeuvres)
Organophosphate poisoning	**Atropine + pralidoxime**
Status epilepticus	**Lorazepam IV** (diazepam/ midazolam)
Eclampsia	**Magnesium sulphate**
PPH	**Oxytocin**
Hypertensive emergency	**Labetalol / nitroprusside / nicardipine**
HTN in pregnancy	**Labetalol / methyldopa / nifedipine**
Acute pulmonary oedema	**Furosemide + nitrates + O_2**
MI (thrombolysis)	**Tenecteplase / streptokinase**
DVT/ PE	**Heparin/ LMWH → warfarin/ DOAC**
Absence seizures	**Ethosuximide / valproate**
Trigeminal neuralgia	**Carbamazepine**
Myasthenia gravis	**Pyridostigmine**
Bipolar disorder	**Lithium**
Parkinsonism	**Levodopa + carbidopa**
Glaucoma (acute)	**Pilocarpine + acetazolamide + mannitol**
Typhoid	**Ceftriaxone / azithromycin**
Scrub typhus	**Doxycycline**
Amoebiasis	**Metronidazole**
MRSA	**Vancomycin**
Syphilis	**Benzathine penicillin**
Vivax malaria	**Chloroquine + primaquine**
Severe malaria	**Injectable artesunate**
Herpes	**Acyclovir**
Systemic fungal	**Amphotericin B**

Condition	DOC
Scabies	**Permethrin 5 %**
Hyperthyroidism	**Carbimazole** (PTU – 1st trimester/ storm)
Diabetes insipidus	**Desmopressin**
T2DM first-line	**Metformin**
DKA	**IV regular insulin + fluids**
Gestational DM	**Insulin**
Acute gout	**Colchicine / NSAIDs**
Chemo vomiting	**Ondansetron**
Hepatic encephalopathy	**Lactulose (+ rifaximin)**
Malignant hyperthermia	**Dantrolene**

10.3 Famous Side Effects (One Drug – One Clue)

Clue	Drug
Dry cough, angioedema	**ACE inhibitors**
Ankle oedema	**Amlodipine**
Gum hyperplasia	**Phenytoin, cyclosporine, nifedipine**
Gray baby	**Chloramphenicol**
Red man	**Vancomycin**
Orange urine	**Rifampicin**
Optic neuritis	**Ethambutol**
Peripheral neuropathy	**INH, vincristine**
Ototoxicity	**Aminoglycosides, furosemide (fast IV), cisplatin**
Tendon rupture	**Fluoroquinolones**
Teeth staining	**Tetracyclines**
Haemorrhagic cystitis	**Cyclophosphamide**
Cardiotoxicity	**Doxorubicin**
Pulmonary fibrosis	**Bleomycin, amiodarone, busulfan**
Yellow-green vision	**Digoxin**
Thyroid dysfunction, blue skin	**Amiodarone**
Lupus-like	**Hydralazine, procainamide, INH**
Agranulocytosis	**Clozapine, carbimazole**
Aplastic anaemia	**Chloramphenicol**
Disulfiram-like reaction	**Metronidazole**
Cinchonism	**Quinine**

Clue	Drug
Lactic acidosis	**Metformin**
Reye's syndrome	**Aspirin (children)**
Hepatotoxicity (overdose)	**Paracetamol**
Muscle pain (myopathy)	**Statins**
Hypokalaemia	**Loop & thiazide diuretics**
Hyperkalaemia	**ACEI, spironolactone**
Gynaecomastia	**Spironolactone, cimetidine**
Tardive dyskinesia	**Antipsychotics, metoclopramide**
Oral thrush	**Inhaled steroids**
Hypoglycaemia	**Insulin, sulfonylureas**

10.4 Narrow Therapeutic Index & Monitoring

Drug	Therapeutic level / monitor
Digoxin	**0.5–2 ng/ml**; apex pulse (hold < 60), K^+
Lithium	**0.6–1.2 mEq/L**; toxic > 1.5
Phenytoin	**10–20 µg/ml**
Theophylline	**10–20 µg/ml**
Warfarin	**INR 2–3** (2.5–3.5 mechanical valves)
Heparin (UFH)	**aPTT 1.5–2.5 × control**
Aminoglycosides / vancomycin	Peak & trough levels, creatinine
Carbamazepine / valproate	Levels, LFT, CBC
Cyclosporine / tacrolimus	Trough levels
Insulin	Blood glucose
$MgSO_4$	**4–7 mEq/L**; knee jerk, RR ≥ 12, urine ≥ 30 ml/hr
Potassium (IV)	Never IV push; ≤ 10 mEq/hr (peripheral)

- **High-alert drugs (ISMP)** – **insulin, heparin/ anticoagulants, KCl concentrate, opioids, chemotherapy, NM blockers, hypertonic saline (3 %), $MgSO_4$, IV sedatives, oxytocin (IV), inotropes** – **independent double-check**, labelled & segregated.
- **LASA (look-alike sound-alike)** – tall-man lettering (e.g. hydrOXYzine/ hydrALAZINE).

10.5 Urine / Stool / Body Colour Changes

Colour	Drug
Orange-red	**Rifampicin**, phenazopyridine, sulfasalazine (orange-yellow)
Red	**Doxorubicin**, phenytoin (pink-red)
Brown/ dark	**Metronidazole, nitrofurantoin**, levodopa, methyldopa (darkens on standing)

Colour	Drug
Blue-green	**Methylene blue**, amitriptyline, propofol (green)
Black stools	**Iron, bismuth**
Skin pigmentation	**Clofazimine (red-brown)**, amiodarone (blue-grey), bleomycin, minocycline

10.6 150 Rapid-Fire One-Liners

150 one-liners

1. Father of pharmacology – **Oswald Schmiedeberg**; father of modern pharmacology in India – **R. N. Chopra**.
2. Pharmacokinetics – **what body does to drug (ADME)**.
3. Pharmacodynamics – **what drug does to body**.
4. First-pass metabolism avoided by – **sublingual, IV, transdermal, rectal (partly)**.
5. Bioavailability of IV – **100 %**.
6. Main site of metabolism – **liver**; excretion – **kidney**.
7. Half-life – **time for plasma concentration to fall by 50 %**.
8. Steady state – **4–5 half-lives**.
9. Loading dose – **to reach steady state quickly**.
10. Therapeutic index – **TD50/ ED50**; narrow – **digoxin, lithium, warfarin**.
11. Enzyme inducers – **rifampicin, phenytoin, carbamazepine, phenobarbitone, alcohol (chronic), smoking**.
12. Enzyme inhibitors – **cimetidine, erythromycin, ketoconazole, ciprofloxacin, valproate, grapefruit**.
13. Tachyphylaxis – **rapid tolerance (ephedrine, nitrates)**.
14. Idiosyncrasy – **genetically determined abnormal response**.
15. Type I hypersensitivity – **anaphylaxis (IgE)**.
16. Placebo – **inert substance**.
17. Orphan drugs – **for rare diseases**.
18. Schedule H – **prescription-only**; Schedule X – **narcotics/ psychotropics (strict)**.
19. Drugs & Cosmetics Act – **1940**; NDPS Act – **1985**.
20. Pharmacovigilance (India) – **PvPI 2010, IPC Ghaziabad**.
21. Rights of medication – **right patient, drug, dose, route, time + documentation**.
22. Patient identification – **2 identifiers**.
23. IM site of choice adults – **ventrogluteal**; infants – **vastus lateralis**.
24. Deltoid max volume – **1 ml (≤ 2 ml)**.
25. Z-track – **iron dextran**.
26. Intradermal angle – **10–15°**; SC – **45–90°**; IM – **90°**.
27. BCG/ Mantoux – **intradermal**.
28. Insulin & heparin – **subcutaneous**.
29. Heparin site – **abdomen, don't aspirate or massage**.
30. Drops/min = **volume × drop factor / time (min)**.
31. Macro-drip – **15–20 drops/ml**; micro-drip – **60 drops/ml**.
32. 1 grain – **60–65 mg**; 1 tsp – **5 ml**; 1 tbsp – **15 ml**.
33. Young's rule – **age/(age+12) × adult dose**; Clark's – **weight (lb)/150 × adult dose**.
34. Ophthalmic drop – **lower conjunctival sac; press inner canthus 1–2 min**.
35. Ear drops – adult **pinna up & back**; child < 3 yrs **down & back**.
36. Sublingual nitroglycerin – **max 3 tabs, 5 min apart**; store **dark glass bottle**.
37. Cholinergic crisis – **SLUDGE**.

38. Atropine toxicity – **"dry as bone, red as beet, hot as hare, blind as bat, mad as hatter"**.
39. Atropine contraindicated in – **narrow-angle glaucoma, BPH**.
40. Physostigmine – **anticholinergic poisoning**.
41. Neostigmine – **myasthenia, reversal of NM block, post-op urinary retention**.
42. Edrophonium (Tensilon) test – **myasthenia gravis**.
43. Adrenaline in anaphylaxis – **0.5 mg IM thigh (0.01 mg/kg child)**.
44. Dopamine low dose – **renal vasodilation**.
45. Dobutamine – **cardiogenic shock (β1)**.
46. Noradrenaline – **septic shock (first-line)**; extravasation – **phentolamine**.
47. β-blocker contraindicated in – **asthma, heart block, bradycardia**.
48. β-blockers mask – **hypoglycaemia**.
49. Propranolol – **non-selective; migraine, thyrotoxicosis, varices**.
50. ACE inhibitor – **cough (bradykinin)**; contraindicated – **pregnancy, bilateral RAS**.
51. First-dose hypotension – **prazosin, ACEI**.
52. Rebound hypertension – **clonidine (abrupt stop)**.
53. Hydralazine – **lupus-like, reflex tachycardia**.
54. Nitroprusside – **cyanide toxicity; protect from light**.
55. Nitrate tolerance – **nitrate-free interval 10–12 hrs**.
56. Digoxin – **check apex 1 min; hold < 60/min (adult)**.
57. Digoxin toxicity ↑ by – **hypokalaemia**.
58. Digoxin toxicity first sign – **anorexia, nausea, vomiting**.
59. Amiodarone – **pulmonary fibrosis, thyroid, corneal deposits**.
60. Adenosine – **very short half-life (< 10 sec), rapid IV push + flush**.
61. Furosemide – **loop of Henle; hypokalaemia, ototoxicity**.
62. Spironolactone – **K-sparing, hyperkalaemia, gynaecomastia**.
63. Mannitol – **cerebral oedema, glaucoma; check crystals**.
64. Acetazolamide – **glaucoma, altitude sickness**.
65. Diuretic best given – **morning**.
66. Heparin – **monitor aPTT; antidote protamine**.
67. Warfarin – **INR; antidote vitamin K; keep vitamin K (green leafy) intake consistent**.
68. Warfarin onset – **3–5 days (overlap with heparin)**.
69. Aspirin low dose – **antiplatelet (75–150 mg)**.
70. Clopidogrel – **P2Y12 inhibitor**.
71. Streptokinase – **antigenic, not repeated (allergy)**.
72. Statin – **night (simvastatin)**; side effect **myopathy, rhabdomyolysis**.
73. Iron – **empty stomach with vitamin C; black stools; straw for liquid**.
74. Iron absorption ↓ by – **tea, milk, antacids, calcium**.
75. KCl IV – **never bolus; dilute; max 10–20 mEq/hr**.
76. Hypertonic saline – **3 %**; isotonic – **0.9 % NS**.
77. Blood transfusion – **within 30 min of issue; finish in 4 hrs; only NS**.
78. Most common transfusion reaction – **febrile non-haemolytic**.
79. Most dangerous transfusion reaction – **ABO incompatibility (acute haemolytic)**.
80. Salbutamol side effects – **tremor, tachycardia, hypokalaemia**.

81. ICS – **rinse mouth (thrush)**.
82. Theophylline toxicity – **seizures, arrhythmias**.
83. Cromolyn – **prophylaxis only**.
84. PPI – **30–60 min before breakfast**.
85. Misoprostol – **NSAID ulcer prevention, contraindicated pregnancy**.
86. Ondansetron – **5-HT3 antagonist**.
87. Metoclopramide – **EPS, acute dystonia**.
88. Lactulose – **hepatic encephalopathy**.
89. Liquid paraffin – **lipoid pneumonia**.
90. Isabgol – **bulk laxative; plenty of water**.
91. Morphine – **respiratory depression (most dangerous), constipation (no tolerance), miosis**.
92. Hold morphine if – **RR < 12/min**.
93. Naloxone – **opioid antagonist (short acting – re-dose)**.
94. Pethidine – **avoid with MAOI; norpethidine seizures**.
95. Paracetamol max – **4 g/day (adult)**; antidote – **NAC**.
96. Aspirin overdose – **tinnitus, respiratory alkalosis → metabolic acidosis**.
97. NSAIDs – **gastric ulcer, renal, ↑ BP**.
98. Phenytoin – **gum hyperplasia, hirsutism, nystagmus; IV in NS only, ≤ 50 mg/min**.
99. Valproate – **hepatotoxicity, NTD, weight gain**.
100. Carbamazepine – **SJS (HLA-B*1502), hyponatraemia**.
101. Levodopa + **carbidopa** (↓ peripheral breakdown); avoid **pyridoxine, high-protein meals**.
102. On-off phenomenon – **levodopa**.
103. Diazepam IV – **slowly, not diluted (precipitates)**.
104. Flumazenil – **BZD antagonist**.
105. Lithium toxicity ↑ by – **dehydration, low salt, thiazides, NSAIDs**.
106. MAOI + cheese – **hypertensive crisis**.
107. NMS – **fever, rigidity, ↑ CPK; dantrolene/ bromocriptine**.
108. Clozapine – **agranulocytosis (weekly WBC)**.
109. Succinylcholine – **depolarising; hyperkalaemia, malignant hyperthermia**.
110. Malignant hyperthermia – **dantrolene**.
111. Ketamine – **dissociative; ↑ BP; emergence reactions**.
112. Propofol – **white milky, lipid; pain on injection; propofol infusion syndrome**.
113. Lignocaine max – **4.5 mg/kg (7 with adrenaline)**.
114. Adrenaline with LA contraindicated – **fingers, toes, penis, nose, ear**.
115. Spinal anaesthesia complication – **hypotension, post-dural puncture headache (lie flat, fluids)**.
116. Penicillin – **anaphylaxis; test dose**.
117. Benzathine penicillin – **RF prophylaxis every 21 days**.
118. Aminoglycosides – **nephro- & ototoxic**.
119. Tetracycline – **teeth staining; avoid < 8 yrs**.
120. Chloramphenicol – **gray baby**.
121. Fluoroquinolones – **tendon rupture**.
122. Cotrimoxazole – **crystalluria; PCP prophylaxis**.
123. Metronidazole – **metallic taste; avoid alcohol**.

124. Vancomycin – **red man; infuse over 60 min**.
125. INH – **pyridoxine**; rifampicin – **orange urine, OCP failure**.
126. Ethambutol – **colour vision**; pyrazinamide – **gout**.
127. Leprosy MDT – **rifampicin, clofazimine, dapsone**.
128. Amphotericin B – **nephrotoxic; 5 % dextrose**.
129. Nystatin – **oral thrush (swish & swallow)**.
130. Acyclovir – **herpes; hydrate**.
131. HIV first-line – **TLD**.
132. PEP – **within 72 hrs, 28 days**.
133. Primaquine – **G6PD haemolysis**.
134. Albendazole – **400 mg single dose**.
135. Insulin – **clear before cloudy**.
136. Only IV insulin – **regular**.
137. Glargine – **peakless**.
138. Hypoglycaemia – **< 70 mg/dl; 15 g rule**.
139. Metformin – **lactic acidosis; B12 ↓**.
140. Levothyroxine – **empty stomach morning**.
141. Carbimazole – **agranulocytosis**.
142. Steroids – **morning; taper**.
143. Oxytocin – **water intoxication**.
144. Methylergometrine – **contraindicated in HTN**.
145. Cyclophosphamide – **MESNA**.
146. Methotrexate – **leucovorin; weekly (RA)**.
147. Vincristine – **never intrathecal**.
148. Iron poisoning – **desferrioxamine**.
149. Pregnancy category X – **isotretinoin, thalidomide, statins**.
150. Elderly – **start low, go slow**.



Exam strategy – Asha Didi ki last advice

- GSSSB Staff Nurse paper me **Pharmacology se ~8–12 direct questions** aate hain, aur MSN/ OBG/ Paediatrics ke questions me bhi drugs chhupe hote hain.
- Sabse zyada pooche jaate hain – **antidotes, side effects (one drug – one clue), drug of choice, dose calculation (drops/min), insulin, digoxin, heparin/ warfarin, $MgSO_4$, oxytocin, emergency drugs, IM sites aur rights of medication**.
- **Calculation** wale questions me formula ek baar likh kar solve karo – units (mg/ mcg/ ml) zaroor check karo.
- "Hold the drug" wale questions – **digoxin (pulse < 60), β-blocker (pulse/ BP low), morphine (RR < 12), $MgSO_4$ (reflex absent/ RR < 12/ urine < 30 ml/hr)**.
- MCQ Bank PDF ke saare questions **2 baar** solve karo aur galat hue questions ko notebook me likho.

— Pharmacology · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —